Operating Manual

DURAG

D-LX 110/710

Compact Flame Monitor

Before starting any work please read the operating manual!

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DURAG

Operating manual Compact Flame Monitor D-LX 110/710

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Contents

1 General information

1.1 Information on this manual

The manual provides important information on handling the product. Compliance with all safety instructions and work instructions is a precondition for safe work.

If anything is unclear:

contact the manufacturer! Get answers to questions.

The manual forms part of the product:

1. Store the manual in the immediate vicinity of the place of use so that it is available at all times!
2. If you hand the device over to a third party, the operating manual must be handed over as well!

The manual

- Relates to the complete system and all variants, even if you have not purchased some of the individual components.
- Relates to the device version at the time the manual was created.
- Contains illustrations which may differ from the actual appearance due to further technical developments or for reasons of clarity.
- Indicates all dimensions in mm (any exceptions are marked accordingly).

Warning

Warning notices are marked with symbols in this manual. These warning notices are introduced by signal words that express the degree of danger.

DANGER

... indicates an immediately dangerous situation that will lead to death or serious injury if not prevented.

WARNING

... indicates a possibly dangerous situation that can lead to death or serious injury if not prevented.

CAUTION

... indicates a possibly dangerous situation that can lead to minor or slight injury if not prevented.

NOTICE

... indicates a situation that can lead to material damage and environmental damage if not prevented.

1.2 Explanation of symbols

The operating manual applies for all compact flame monitor models (in short: flame monitor) D-LX 110/710. Certain sections do however apply only for specific models, and are labelled accordingly. Information on your device's model (e.g. housing M4 or M5) can be found on the type label (see section 8.9 Type label [▸ 38]). Explanations on the type code can be found in section8.8 Type code [▸ 37].

Possible housing

M4	M5 with	
	Plastic plug-in cover	Metal plug-in cover
		

Applies to all devices with housing M4 (suitable for operation in zone 1/21 and Div. 1 of hazardous areas)

Only applies to D-LX 710

Applies to D-LX 110 with housing M5 (aluminium)

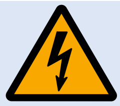
Applies to all devices with housing M4 (suitable for operation in zone 1/21 of hazardous areas) and the devices with housing M5 that offer explosion protection: Models D-LX*M5/86Ex/* and D LX*M5/87Ex/*. Device models D-LX*M5/87Ex/* are suitable or operation in zone 2/22 of hazardous areas (see also Section 8.8 Type code [▸ 37]).

Applies to all the devices with housing M5 that offer explosion protection: Models D-LX*M5/86Ex/* and D LX*M5/87Ex/*. Device models D-LX*M5/87Ex/* are suitable or operation in zone 2/22 of hazardous areas (see also Section 8.8 Type code [▸ 37]).

Applies to all the devices with housing M5 that offer **no** explosion protection: Models D-LX*M5/0000/*. These are **not** suitable for operation in hazardous areas!

Specific safety instructions

To draw attention to specific hazards, warning notices and signal words are used in conjunction with the following symbols:

	General warning symbol		Electric power
	Hot surface		Explosive atmosphere
	Electrostatic sensitive devices (ESD)		

Other symbols used

This symbol indicates the required special tool and the particular technical equipment.

1.3 Customer Service

If you require support or further technical information, DURAG Service is at your disposal. For contacts and manufacturer addresses, visit **www.durag.com**.

Our employees are also always interested in hearing about your experience with the use of our products. Your experience provides us with important information for the further development and improvement of our products.

1.4 Spare parts

WARNING

Risk of injury due to wrong spare parts!

Incorrect or defective spare parts can result in damage or malfunctions and may also impair safety.

- Only use original spare parts of the manufacturer.
- Procure spare parts through authorised dealers or directly from the manufacturer.

2 Safety

Health hazards or material damage may arise due to improper use or handling. Therefore, please observe the following safety information as well as the warning notices in the following sections.

- **The flame monitor device has a type approval. Any intervention or modification to the device will void the type approval. Repairs must only be performed by the manufacturer or persons that it has authorised to do so. This is the only way of ensuring compliance with the specifications for the type approval.**
- When preparing and performing work: Follow the statutory regulations valid for the plant and the corresponding technical rules. Observe national safety and accident prevention regulations.
- Provide a sufficient amount of suitable protective equipment and personal protective equipment. Use this equipment in accordance with the respective potential danger.
- Changes from normal operation indicate functional impairments and should be taken seriously. Pay attention to the following in this respect:
  - Formation of smoke or unusual odours
  - Unusual noises caused by and during operation of the device
  - Overheating in system components
- The operating manual forms part of the product: Store in the immediate vicinity of the device.
- Only operate the device in perfect condition, according to the performance data and in compliance with the safety instructions.
- Only operate the devices as a whole and the individual components in their original designs. Do not carry out any conversions.

2.1 Hazards due to electric current

DANGER

Risk to life due to electric current!

There is an immediate risk to life when touching parts carrying live voltage. Damage to the insulation or individual components can present a risk to life.

- ▶ Only have work on electric systems performed by specialised electricians.
- ▶ Disconnect the flame monitor, check that no voltage is present and secure against reactivation before removing the cover.
- ▶ In the event of damage to the insulation, turn off the power supply immediately and initiate repairs.
- ▶ Keep moisture away from live parts. This may lead to a short-circuit.

To avoid hazards:

- Only connect the flame monitor and contact circuit (operational-readiness and flame contact) to permissible supply voltages.
- Always use a protective extra-low voltage (PELV) system to power the flame monitor! This is the only way of ensuring safe adjustment during operation!

NOTICE	**Damage to electronic components due to electrostatic discharge (ESD)** Electronic components are becoming smaller and smaller and more and more complex. Their susceptibility to damage from electrostatic discharge is increased accordingly. Therefore: ▸ To protect the components, measures must be undertaken to prevent electrostatic discharge during all work performed at the open device (ESD protection). ▸ To prevent static charges building up on the human body, service employees can for example be equipped with a personal earthing system.

2.2 Risk of burns due to hot surfaces

The flame monitor (including accessories) is installed directly onto the burner. The components may become hot during operation. In plants with a low hazard potential (ambient pressure, low temperatures, no toxic gases, no health risks), installation/removal may take place during plant operation, if this is permitted by the operating company's applicable regulations.

CAUTION	**Risk of burns due to hot surfaces!** Contact with hot components may cause burns. Therefore: ▸ Wear protective work clothing and protective gloves when working near hot components. ▸ Before performing any work, make sure that all components have cooled to the ambient temperature.

2.3 Damage due to failure of the purge air

The purge air serves to protect the system fitted onto the boiler. The purge air shields components from hot and/or aggressive gases and dusts.

NOTICE	**The system may be damaged and contaminated due to failure of the purge air!** If the purge air fails, the lens may quickly become contaminated and the flame monitor may overheat due to a build-up of heat and be destroyed. Therefore: Make sure that ▸ the purge air supply is operating reliably and smoothly (even when the system is stopped). ▸ you immediately detect a failure of the purge air. ▸ the flame monitor is protected by a valve or dismantling process in the event of a purge air failure.

2.4 Qualified personnel

WARNING	**Danger of injury due to insufficient qualification!** Incorrect use can result in serious personal injury and material damage. Therefore: ▸ Ensure that work is only ever performed by suitably qualified specialised personnel.

This operating manual assumes that the personnel performing the work have the necessary training and knowledge. Only personnel who have this knowledge are considered to be qualified and authorised in this operating manual.

- **Specialised personnel** are those who because of their specialist training, knowledge and experience, coupled with knowledge of the applicable regulations, are in a position to perform the work assigned to them and make independent judgements of the potential hazards.
- An **electrician** is an individual who, because of his or her specialist training, knowledge and experience, coupled with knowledge of the applicable standards and regulations, is in a position to perform work on electrical equipment and make independent judgements of the potential hazards. Electricians are specifically trained for the working environment in which they are employed.
- **Electrically trained persons** are those who have been instructed by a specialised electrician on the work assigned to them and the potential hazards involved with improper conduct, and on the necessary protective equipment and protective measures.
- **Knowledge of explosion protection** is required by all persons carrying out work on the flame monitor in any kind of hazardous area. This requires knowledge of the detection and prevention of explosion hazards in accordance with the explosion protection document. Employees must know how to avoid acute risks and where work can safely be performed on the devices.
  The specialist knowledge on explosion protection must comply with the national regulations at the place of installation of the device (ATEX, FM, CSA, ...). The permit and qualification or certification required to work in areas at risk of explosion are regulated in national regulations or laws.

The operating company must ensure that the aforementioned specialised personnel are given up-to-date instructions on the following:

- Precise information on the operating risks and their prevention.
- Knowledge of system conditions, applicable standards, regulations, directives, operating instructions and accident prevention regulations in the context of the work assigned to them.
- **Sufficient knowledge of the flame monitor DURAG offers appropriate training courses in order to acquire specialist knowledge of the device. Information on these is available online on the DURAG homepage or by telephone (see the manufacturer's address on page 2).**

Specialised personnel are in principle always required for all tasks. Any additional necessary qualifications are specified at the start of the respective sections.

2.5 Personal protective equipment

It may be necessary to wear personal protective equipment while working to minimise the health risks. The operator is responsible for creating detailed specifications based on the plant-specific potential danger.

3 Functional safety

This section relates only to those devices and components that perform safety functions as parts of a system's safeguarding equipment and within the context of functional safety, and that have been inspected in order to work out the Safety Integrity Level (SIL). Within product family D-LX 110/710, these are product versions D-LX *** ***-*1/*. The only output signal that is relevant to safety within the sense of functional safety is the flame ON contact.

3.1 Relevant standard

The relevant standard is

- DIN EN 13611 Appendix J (edition as per the relevant SIL document) Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems

The term definitions and requirements set out in this standard are applied as part of this section without being repeated.

3.2 Functions of the flame monitor

The flame monitor is used for the monitoring of industrial burners. When a flame is present, the fuel being transported into the combustion chamber is consumed by means of a reaction with oxygen and no combustible materials can accumulate. If the flame were to go out, however, any fuel still introduced into the combustion chamber would not be consumed, and together with the air in the chamber could form an explosive mixture that would react immediately on contact with a source of ignition.

The flame monitor evaluates a suitable physical signal from the flame in order to establish whether or not a flame is present. The measurement signal acquired from the physical signal is compared with the threshold values defined by the operator during commissioning of the device. If the measured signal is higher than the threshold value, this indicates that a flame is present. If it is lower than the threshold value, the flame signal disappears. The setting to be selected is described in section 14 Setup [▶ 62].

The flame monitor's operating function consists of the continual monitoring for the presence of a flame. The presence of a flame is indicated by the flame relay having a closed flame ON contact. If no flame is present, this is indicated by an open flame relay. For more information on this, see also section 9 Functional description [▶ 39].

The safety function of the flame monitor consists of indicating at the flame relay the loss of a flame during regular operation of the burner (e.g. flame rupture), at the latest following a pre-defined Flame Failure Detection Time (FFDT) specified by the operator within the context of the applicable standards. The loss of a flame is indicated by the opening of the relay contact. This must result in the supply of fuel being interrupted by the operator. The flame monitor is part of the functional safety chain (flame signal → flame monitor → safety controller → fuel valves → fuel flow) and so must be included in an overall safety validation process that takes the entire safety chain into account.

3.3 General applicable parameters for the flame monitor

Regardless of its operating mode, the compact flame monitor will always have the following applicable parameters:

Evaluated as per	DIN EN 13611 Appendix J
Suitable for use at up to	SIL3

3.4 Operation with low and high demand modes

A safety-relevant system can be designed for a low or high demand mode.

- Operation with low demand mode
  The rate of demand for the safety-related system does not exceed once per year.
- Operation with high demand mode
  The rate of demand for the safety-related system does exceed once per year.

The flame monitors being considered here can only be used in the "high demand mode" operating mode, taking into account EN 13611, Appendix J.

3.5 Parameters in operation with high demand mode

Device model	UA/UAF	IG/IS	UL
$\lambda_{sd} + \lambda_{su}$	1.84×10^{-6} / h	1.46×10^{-6} / h	4.89×10^{-5} / h
λ_{dd}	1.83×10^{-6} / h	1.81×10^{-6} / h	4.13×10^{-6} / h
λ_{du}	4.32×10^{-8} / h	4.34×10^{-8} / h	6.64×10^{-8} / h
SFF	98.8 %	98.7 %	99.9 %
PFH_D	2.24×10^{-8} / h	2.25×10^{-8} / h	4.56×10^{-8} / h
DC_{AVG}	97.7 %	97.7 %	98.4 %
Proof test interval	10 years		

The proof test interval is the same as the 10-year planned service life specified based on product standard EN 298. Proof testing can therefore be omitted within this period.

4 Explosion protection

This section affects only those flame monitors with housings M4 (suitable for operation in zone 1/21 and Div. 1 of hazardous areas) and the devices with housing M5 that offer explosion protection: Models D-LX*/86Ex/* (suitable for operation in Div. 2 of potentially explosive areas) and D-LX*/87Ex/* (suitable for operation in zone 2/22 of potentially explosive areas).

Information on the housing is provided as part of the model designation on the type label (see section 8.9 Type label [▶ 38]). Explanations on the type code can be found in section8.8 Type code [▶ 37]. Information on the explosion protection can be found on the Ex type label (see section 4.8 Ex type label [▶ 23]).

4.1 Basics

In many areas of industry, flammable gases, vapours, mists or dust are produced as a result of production, processing, transport and storage. These combustible materials can form explosive atmospheres with oxygen, for instance from the air. Should these ignite, for example due to electrical sparks, serious personal injury and material damage may result. The company which operates the devices is responsible for preventing any risk of explosion and guaranteeing the required level of safety.

The product therefore needs to be structurally designed in accordance with the latest knowledge regarding explosion protection and manufactured to an appropriate level of quality. In addition, professional installation and preventative maintenance of all relevant components are required.

Devices for hazardous areas are relevant in terms of safety and are subject to a particular duty of care. Expert knowledge of explosion protection (corresponding to the national regulations) is a prerequisite for any planning or work. This also applies to the choice of suitable ignition protection categories and the classification of hazardous areas into zones, as well as the correct assembly, commissioning, repair or exchange of devices and accessories.

The basic principles for explosion protection are the same all over the world. In the Member States of the European Union, explosion protection is governed by the ATEX guidelines (French ATmosphère EXplosible). The guidelines are implemented through harmonised European standards which to a large extent correspond to the IECEx standards (International Electrotechnical Commission Explosive). Differences for instance arise with respect to North American explosion protection.

4.2 Classification of the equipment groups into categories

Device group I covers explosion-proof devices used in areas with potentially explosives in the mining industry.

Device group II covers all other explosion-proof devices.

The flame monitor is only suitable for hazardous areas in device group II.

Device group II is divided into categories 1, 2 and 3, with decreasing levels of safety.

4.3 Zone allocation as per ATEX

The areas where potentially explosive mixtures can develop are in principle divided into zones with different safety levels. Distinctions are made between gaseous and dust-based atmospheres, and also based on the expected frequency of the development of explosive mixtures.

Zone	Potentially explosive atmosphere	Suitable devices
Zone 1	Frequent (2), gas (G)	II 2 G
Zone 2	Infrequent (3), gas (G)	II 2 G, II 3 G
Zone 21	Frequent (2), dust (D)	II 2 D
Zone 22	Infrequent (3), dust (D)	II 2 D, II 3 D

Table 4.1: Zones for the use of explosion-proof flame monitors

4.4 Temperature classes

Flammable gases and dust are divided into temperature classes based on their ignition temperature. The ignition temperature of a potentially explosive gas or dust atmosphere gives the maximum surface temperature that may be reached here by the devices being used.

Electrical operating materials that may be used in hazardous areas indicate the temperature classes of the gases for which they are suitable. The temperature class specified on the device must be the same or higher than that of the potentially explosive atmosphere.

The surface temperature of the devices is influenced by the ambient temperature. Therefore, the ATEX ambient temperature (Ta) specified on the device (type label) must also be complied with.

4.5 EPL Equipment Protection Level

The inclusion of Equipment Protection Levels (EPLs) for explosion protected operating equipment represents a new alternative to the concept of zones for the choice of devices. It is mentioned here for information purposes. The risk of internal ignition can be assessed in a better and more flexible way by means of a risk assessment. Each zone can be assigned a precise equipment protection level without an additional risk assessment.

The following assignments of DURAG devices are important:

Zone	EPL
Zone 1	Gb
Zone 2	Gc
Zone 21	Db
Zone 22	Dc

Table 4.2: EPL of DURAG devices

4.6 Ignition protection categories

The highest degree of safety when using flammable substances is achieved if the development of a potentially explosive atmosphere is prevented (primary explosion protection).

In areas where potentially explosive atmospheres need to be expected despite all necessary measures being taken, only explosion-protected operating materials may be used.

The ignition protection type refers to different design principles that are used to prevent a potentially explosive atmosphere and an ignition source from existing simultaneously.

Explosion protection is achieved by means of a gas-proof/dust-proof device encapsulation. This means that potential ignition is prevented, because the potentially explosive atmosphere does not reach the ignition source.

There is a range of different ignition protection types with various safety levels (device categories) that the device can comply with. All ignition protection types for the assemblies used are standardised, and the protection types within the individual categories should be considered as being equivalent in terms of safety.

4.6.1 Pressure-proof encapsulation "Ex d"

This ignition protection type is used for device group and category II 2 G (zone 1). Housings for devices of this ignition protection type are so robust that they can withstand internal explosions without damage. Design-based measures prevent the explosion from spreading to a potentially explosive atmosphere outside the device (tertiary explosion protection)

Unavoidable gaps from the inside to the exterior space (e.g. between the housing body and sealing cover) may exist. They are so small that the flame front of an internal explosion is extinguished if it escapes.

The gap surfaces must not be cleaned with mechanical devices such as a wire brush or abrasives. Even small scratches on the gap surfaces represent hazardous damage that can lead to ignition.

Components that hold these housings together (e.g. screws and cable glands) are safety-relevant. For sufficient safety of cable glands in the Ex d space at least five turns of the thread must mesh perfectly. Safety-relevant components must only be replaced with approved spare parts. Only these spare parts have been demonstrated to be able to withstand the strains they are subjected to.

Ignition-proof gaps must only be repaired according to the design specifications of the manufacturer. Repairs according to the values of Table 1 or 2 of EN 60079-1 are not permissible.

WARNING

EX

Risk of explosion due to the operation of a flame monitor with a damaged Ex d housing!

The Ex d housing contains sources of ignition. The operation of a damaged or modified Ex d housing in a potentially explosive atmosphere can lead to an explosion. Therefore:

In order not to change the gap dimensions in a way that makes them impermissible:

- Do not make any retroactive changes to the housing
- Have damaged housings or damaged ignition-proof gap surfaces replaced by a certified expert company.
- Only use original screws for the closures
- Only use permissible cable glands so as to guarantee the explosion protection. With the cables used, observe the cable diameter clamping area permissible for the cable gland.
- Only use cables that are suitable for use in the hazardous area (installation according to EN 60079-14).

4.6.2 Increased safety "Ex e"

This ignition protection type is also used for **device group and category II 2 G (zone 1)**. In Europe it is used for connection compartments in combination with ignition protection type "Ex d" (flame-proof enclosure). As with the "flame-proof enclosure", an explosive atmosphere inside the Ex e area is expected when the device is located within an area containing flammable substances. With the ignition protection type "Increased safety", design-based and technical measures are used to prevent sparks or electric arcs from occurring in normal operation or in the event of a fault. The emergence of hot surfaces is also prevented. The housings are mechanically sturdy and have a protection class (DIN EN 60529) of at least IP54. This means that ingress of moisture and dirt is only permitted at the limits of the specified protection class.

This protection type ensures that air and creepage distances are not reduced during operation and that no sparks can be generated as a result of short circuits. The contact points for conductors are secured against unintentional loosening.

DANGER

EX

Danger of explosion due to electrical sparks in explosive atmospheres!

Explosions can endanger the health and safety of anyone present and cause considerable material damage.

Therefore:

- Do not disconnect cables in hazardous areas when live.

The dimensions of the conductor cross-sections are designed in a way that prevents overheating as a result of high current densities (observe permissible surface temperatures). The rated values for the terminal blocks apply, in particular the conductor cross-section and torque.

Release the conductor ends sufficiently from the isolation in order to guarantee secure positioning. Wires must not, however, be freely outside of the terminal.

4.6.3 Ignition protection type "Ex n" (gas)

This ignition protection type is used for **device group and category II 3 G (zone 2)**. The operating materials covered by this type are not capable of igniting a potentially explosive atmosphere in normal operation or defined abnormal operating conditions.

The contacts of sparking operating materials are protected in a suitable manner; "Ex nC"

There are encapsulated relays inside the flame monitor housing. These correspond to ignition protection type "Ex nC". The cable gland corresponds to an ignition protection type that can be combined with "Ex n_".

WARNING

Risk of explosion due to incorrect installation!

The use of unsuitable components can lead to explosion during operation. Therefore:

- Only ever use cable glands with ATEX approval.
- Only use cables that are suitable for use in the hazardous area.
- In potentially explosive atmospheres, or atmospheres that are presumed to be potentially explosive:
- Do not remove or replace fuses when under voltage.
- Do not disconnect cables from the terminal strip when under voltage.

4.6.4 Protection in the form of an "Ex t" housing (dust)

This ignition protection type is used for the **device group and category II 2 D (zone 21) and II 3 D (zone 22)**. Devices in this ignition protection type have a dust-proof housing (IP6X) and prevent hot surfaces that could cause explosive dust to ignite. It is therefore important to comply with the permissible ambient temperatures. The through-hole threads on the housing are either self-sealing or have seals to protect them against the ingress of dust.

WARNING	**Risk of explosion due to the operation of a flame monitor with a damaged housing!** The Ex t housing contains sources of ignition. The operation of a damaged or modified Ex t housing in a potentially explosive atmosphere can lead to an explosion. Therefore: ▸ Only use permissible cable glands so as to guarantee the explosion protection. With the cables used, observe the cable diameter clamping area permissible for the cable gland. ▸ Only use cables that are suitable for use in the hazardous area (installation according to EN 60079-14). ▸ Do not make any retroactive changes to the housing. In particular, do not use impermissible surface protection, such as paint (as this will impair the protection against electrostatic charges). ▸ Always comply with the existing IP code (IP6X). ▸ Before closing the housing completely remove any dust that may be present from the housing. ▸ Have damaged housings or seals replaced by a certified expert company.

4.7 Explosion labelling

4.7.1 ATEX Markings

Explosion-proof devices in the European Union must be tested, approved and operated in accordance with ATEX guidelines.

Housing	ATEX labelling	Ambient temperature	Suitable for
M4	II 2 G Ex db IIC T5 Gb	D-LX 110-*** UL-***: -40°C ≤ T_a ≤ 70°C All other: -40°C ≤ T_a ≤ 75°C	Zone 1 and zone 2
	II 2 G Ex db IIC T6 Gb	-40°C ≤ T_a ≤ 70°C	
	II 2 D Ex tb IIIC T100 °C Db	D-LX 110-*** UL-***: -40°C ≤ T_a ≤ 70°C All other: -40°C ≤ T_a ≤ 75°C	Zone 21 and zone 22
	II 2 D Ex tb IIIC T85 °C Db	-40°C ≤ T_a ≤ 70°C	
Ex M5	II 3 G Ex ec nC IIC T4 Gc	-40°C ≤ T_a ≤ 70°C	Zone 2
	II 3 G Ex ec nC IIC T6 Gc	-40°C ≤ T_a ≤ 40°C	
	II 3 D Ex tc IIIC T130 °C Dc	-40°C ≤ T_a ≤ 70°C	Zone 22
	II 3 D Ex tc IIIC T80 °C Dc	-40°C ≤ T_a ≤ 40°C	

EU type examination certificate number	
M4	ATEX: IBExU11ATEX1050 X IECEx: IECEx IBE11.0012X
Ex M5	ATEX: IBExU18ATEXB016 IECEx: IECEx IBE18.0024

4.7.2 Labelling as per NEC 500

The NEC 500 and NEC 505 Standards are decisive in the design approval for explosion-proof devices in North America. Devices with the corresponding labellings are approved for use in so-called *hazardous (classified) locations*. The NEC 500 sets out the following classifications and labellings:

Class

Distinctions are made between devices that can be used in areas with the following different atmospheric properties:

- Class I: flammable gases, vapours or mist
- Class II: flammable dust
- Class III: flammable fibres and lint

Divisions

An additional criterion is the frequency or period of time that the substances are emitted at a dangerous concentration.

- **Division 1**: constant or occasional presence (under normal operating conditions) of the ambient conditions described in the *Classes* section.
- **Division 2**: rare presence, or presence that is unexpected in the particular environment (under normal operating conditions), of the ambient conditions described in the *Classes* section.

Groups

Potentially explosive substances are further classified into Groups. For Class I, gases, vapours and mist can be sub-divided into Groups A, B, C and D. Flammable dust is sub-divided into Groups E, F and G.

Temperature classes

For North American explosion protection, classification into temperature classes is performed similarly to the procedure outlined in section 4.4 Temperature classes [▶ 16]. The maximum surface temperature is classified into Classes T1 to T6, in accordance with NEC 500. This classification corresponds to the ATEX temperature classes, but does have additional sub-groups (e.g. T2A, T2B etc.). The temperature classes specified on the type label for the permissible ambient temperature ranges must always be complied with in order to guarantee the explosion protection.

Zone classification and housing degrees of protection

Classification into zones (as per the ATEX guidelines) has been omitted as per NEC 500.

The housing degree of protection is described in accordance with NEMA (National Electrical Manufacturing Association). The NEMA protection classes are not directly comparable with the European degrees of protection. The grading is based on different environmental influences and criteria in the relevant guidelines.

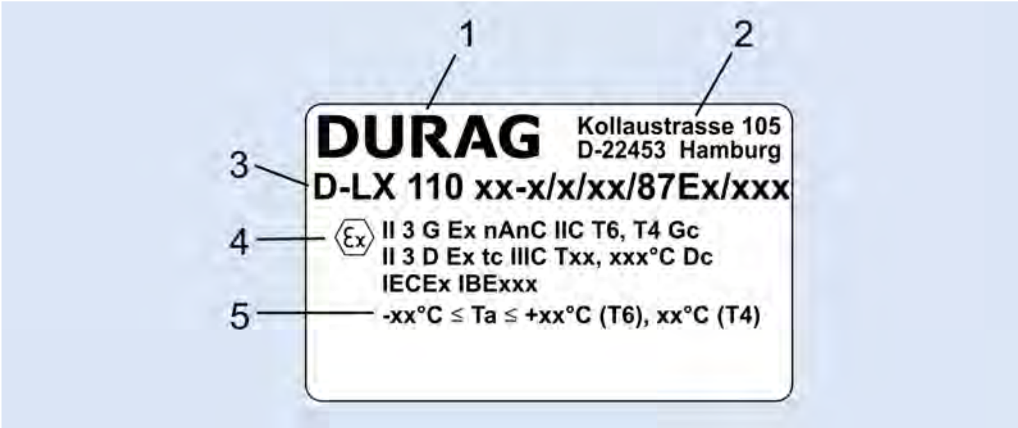
The requirements for the flame monitor housings in terms of protection against damage caused by internal explosions are met through design-based measures. The selection of suitable components also ensures that there is no possibility of incidents occurring in normal operating conditions as a result of sparks either in or on the housing.

Labelling as per NEC 500

Housing	Labelling as per NEC 500	Ambient temperature
M4	Class I, Div. 1, Group A, B, C, D, T5	D-LX 110-*** UL-***: -40°C ≤ T_a ≤ 70°C All others: -40°C ≤ T_a ≤ 75°C
	Class I, Div. 1, Group A, B, C, D, T6	-40°C ≤ T_a ≤ 70°C
	Class II, Div. 1, Group E, F, G, T5; Class III	D-LX 110-*** UL-***: -40°C ≤ T_a ≤ 70°C All others: -40°C ≤ T_a ≤ 75°C
	Class II, Div. 1, Group E, F, G, T6; Class III	-40°C ≤ T_a ≤ 70°C
Ex M5	Class I, Div. 2, Group A, B, C, D, T4	-40°C ≤ T_a ≤ 60°C
	Class I, Div. 2, Group A, B, C, D, T6	-40°C ≤ T_a ≤ 40°C
	Class II, Div. 2, Group E, F, G, T4; Class III	-40°C ≤ T_a ≤ 60°C
	Class II, Div. 2, Group E, F, G, T6; Class III	-40°C ≤ T_a ≤ 40°C

4.8 Ex type label

Flame monitors designed for operation in hazardous areas also have an Ex type label. The approval for use in hazardous areas is only valid in connection with the corresponding labelling on the Ex type label of the flame monitor. Only those flame monitors with the corresponding labelling may be used in a potentially explosive environment.



1	Manufacturer
2	Manufacturer address
3	Device type
4	Ex labelling (see following table for meanings)
5	Ambient temperature

Fig. 4.1: Ex type label (example)

Information	Meaning
II	Device group II = other hazardous areas (not the mining industry)
2 G	Device category 2 G = zone 1 atmosphere (gas)
Ex de	d = flame-proof enclosure, e = increased safety (ignition protection types for electrical operating materials in areas at risk of gas explosions)
IIC	IIC = explosion group for gases (as per IEC, CENELEC and NEC 505; typical gas: hydrogen)
T6	T6 = temperature class for gases (maximum permissible housing surface temperature 85 °C; as per IEC, CENELEC and NEC 505)
IP 66	Protection class 6_ = protection against solid bodies, complete anti-touch protection, protection against the ingress of dust; _6 = protection against the ingress of water during temporary flooding
Gb	Device protection level: Gb corresponds to zone 1

Table 4.3: Ex labelling (example!)

The ambient temperature Ta may vary from the temperature specified on the type label. Explanations on the contents of the type label can be found in section 8.9 Type label [▸ 38].

4.9 Safety with explosion-proof devices

4.9.1 All devices

Devices with approvals for use in hazardous areas must only be installed, put into operation and maintained by qualified personnel with knowledge of explosion protection (see section 2.4 Qualified personnel [▸ 11]). The national assembly and installation regulations apply (e.g. IEC/EN 60079-14, NFPA 70/NEC 500).

WARNING

EX

Danger of explosion from use of devices in explosive atmospheres!

Devices without explosion protection and damaged devices may ignite an explosive atmosphere.

Therefore:

- In hazardous areas, only use device types which are appropriately marked and intended for this environment (see type label).
- Only use undamaged devices.
- Only use approved original spare parts when carrying out repairs.

WARNING

EX

Risk of explosion due to unauthorised technical modifications.

The flame monitor device has a design approval. Any intervention or modification to the device will void the design approval and can lead to an explosion. Therefore:

- Only ever have repairs carried out by the manufacturer or its authorised representatives. This is the only way of ensuring compliance with the specifications for the design approval.

5 Designated use

The flame monitor is a safety device for the monitoring of firing systems with an extremely wide range of fuels and combustion methods, and in both intermittent and continuous operation. The special requirements of DIN EN 50156-2:2015 Section 10.5.5 are also complied with here.

Possible application areas are:

- Large-scale power plants
- Remote heating stations
- Chemical process incinerators
- Thermal exhaust gas incinerators
- Sulphur recovery units (SRUs)
- Rotary kilns
- Refineries etc.

Fault-free and safe operation of the device depends on appropriate transport, correct storage and assembly, as well as careful operation and maintenance by qualified personnel.

The flame monitor must only be used for the designated use described here and only in the design intended for the respective application. The corresponding explosion-protected version must be used in hazardous areas (observe Ex type label).

WARNING

Danger due to incorrect use!

Any use other than or beyond the designated use of the flame monitor can lead to dangerous situations. There is a risk of personal injury and material damage. Therefore:

- Only use the flame monitor in accordance with the information on the type label.
- All instructions and specifications in this operating manual must be strictly complied with.

6 Which flame monitor do I need?

Type	Spectral range (nm)	Suitable for fuels				Features
		Gas	Oil	Coal	Wood	
D–LX 110 UL	185 … 260	++	+			Monitoring of gas and oil flames
D–LX 110 UA D–LX 710 UA	190 … 520	++	++	0		Monitoring of gas and oil flames, even in low-NOx combustion.
D–LX 110 UAF D–LX 710 UAF	280 … 410	+	++			Monitoring of very intensive gas and oil flames, even in low-NOx combustion.
D–LX 110 IS D–LX 710 IS	300 … 1100	!	++	+	+	Monitoring of oil flames, even when additional fuels are also being burned.
D–LX 110 IG D–LX 710 IG	780 … 1800	0	+	++	++	Monitoring of oil and wood dust flames and coal flames.

Table 6.1: Suitability of device variants (spectral ranges)

Symbol guide:

+ The flame monitor is suitable for use with this fuel.

++ The flame monitor is ideally suitable for use with this fuel.

0 The flame monitor is suitable to a limited extent for use with this fuel. The monitoring properties would depend to a large extent on the combustion method.

! The flame monitor is often not permitted for the monitoring of gas flames due to local regulations.

The table is based on years of experience with a range of different combustion plants. A specific flame behaviour due to the combustion method used in an individual case cannot, however, be taken into account. As a result, there may be deviations from this table.

All flame monitor types are offered in the following housings:

Name	Material	Explosion protection
	Aluminium	Suitable for zone 2 and zone 22
		No
	Aluminium	Suitable for zone 1/2 and zone 21/22

Table 6.2: Housing and explosion protection

7 Technical data

7.1 Devices with housing M5

M5 Models D-LX bbb ccc-dd/**M5**/ffff/ggg	
Dimensions	**D-LX 110** Approx. 100 x 100 x 260 mm (with purge air connection) **D-LX 710** Approx. 100 x 100 x 180 mm (no purge air connection)
Weight (without cable)	Approx. 1.3 kg
ATEX labelling	Only a few types offer explosion protection: for more information on this, see section 4.7.1 ATEX Markings [▶ 21]
Ambient temperature	-40 … +75°C, (D-LX 110-*** UL-***: -40 … +70°C), see also section 4.7 Explosion labelling [▶ 21]
Humidity	≤ 95 % relative humidity, non-condensing
Only models D-LX 110***UL-***: Service life of a UV cell	At least 10,000 operating hours but significantly longer in favourable operating conditions; replacement necessary if sensitivity is reduced
Mode	Intermittent operation, continuous operation
Flame Failure Detection Time (FFDT)	1 s (adjustable: 0.5 s, 1 s, 1.5 s, … 5 s); *for information on restrictions with regard to FM and AGA approvals, see section* 14.3 Set flame failure detection time (FFDT) [▶ 67]
Electrical connection	2.0 m cable 1 x 0.75 mm² and 4 x 2 x 0.75 mm² with complete shielding
Housing	Aluminium, painted blue
Protection type (as per EN 60529)	IP 65 *for models ***/MP7* IP 66 *for all other models*
Protection class	III
Operating voltage	24 V⎓ ±20 % PELV (in accordance with IEC 60730-1)
Power consumption	Max. 7 W
Protection	F1= 0.5 A delayed action, MST 250
Analogue output 0/4 ... 20 mA	Load max. 750 Ω
Flame contact	Positively driven Normally open contact: active when flame ON Normally closed contact: active when flame OFF (not safety-relevant)
Ready for operation contact	Normally open contact: active if no fault
Switching capacity of the relay contacts	**Max. 24 V⎓** Min. 10 mA at 10 V⎓ Max. 0.5 A at 24 V⎓ with spark suppression **Max. 50 V~** *for models ***/MP7* /**max. 250 V~** *for all other models* Min. 10 mA at 10 V~ Max. 2.0 A /cos φ = 1.0 (resistive load) Max. 1.0 A /cos φ = 0.6 (inductive load)

Fuse F3	Max. 24 V⎓ 0.5 A with spark suppression: 0.5 A delayed action MST 250 Max. 50 V~ *for models ***/MP7* /max. 250 V~ *for all other models* 2.0 A /cos φ = 1.0 (resistive load): 2 A delayed action MST 250 1.0 A /cos φ = 0.6 (inductive load): 1 A delayed action MST 250 The devices are delivered with F3 = 0.5 A delayed action MST 250 ex-works as standard. Other fuses available on request.

7.2 Devices with housing M4 for zone 1/21

M4 Models D-LX bbb ccc-dd/**M4**/ffff/ggg		
Dimensions	D-LX 110 Ø120 mm, length 310 mm (with purge air connection)	D-LX 710 Ø120 mm, length 230 mm (no purge air connection)
Weight (without cable)	Approx. 3.0 kg	
ATEX labelling	See section 4.7.1 ATEX Markings [▸ 21]	
Ambient temperature		
Humidity	≤ 95% relative humidity, non-condensing	
Only models D-LX 110*****UL**-***: Service life of a UV cell	At least 10,000 operating hours but significantly longer in favourable operating conditions; replacement necessary if sensitivity is reduced	
Mode	Intermittent operation, continuous operation	
Flame Failure Detection Time (FFDT)	1 s (adjustable: 0.5 s, 1 s, 1.5 s, … 5 s); *for information on restrictions with regard to FM and AGA approvals, see section* 14.3 Set flame failure detection time (FFDT) [▸ 67]	
Electrical connection	2.0 m cable 1 x 0.75 mm² and 4 x 2 x 0.75 mm² with complete shielding	
Housing	Aluminium die-casting, ESD paint, blue	
Protection type (as per EN 60529)	IP 66	
Protection class	III	
Operating voltage	24 V⎓ ±20 % PELV (in accordance with IEC 60730-1)	
Power consumption	Max. 7 W	
Protection	F1= 0.5 A delayed action, MST 250	
Analogue output 0/4 ... 20 mA	Load max. 750 Ω	
Flame contact	Positively driven Normally open contact: active when flame ON Normally closed contact: active when flame OFF (not safety-relevant)	
Ready for operation contact	Normally open contact: active if no fault	

Switching capacity of the relay contacts	**Max. 24 V⎓** Min. 10 mA at 10 V⎓ Max. 0.5 A at 24 V⎓ with spark suppression **Max. 250 V~** Min. 10 mA at 10 V~ Max. 2.0 A at 250 V/cos φ = 1.0 (resistive load) Max. 1.0 A at 250 V/cos φ = 0.6 (inductive load)
Fuse F3	**Max. 24 V⎓** 0.5 A with spark suppression: 0.5 A delayed action MST 250 **Max. 250 V~** 2.0 A /cos φ = 1.0 (resistive load): 2 A delayed action MST 250 1.0 A /cos φ = 0.6 (inductive load): 1 A delayed action MST 250 The devices are delivered with F3 = 0.5 A delayed action MST 250 ex-works as standard. Other fuses available on request.

7.3 Purge air and pipe connections

D-LX 110

Purge air Operating pressure Consumption	Approx. 1 … 50 hPa Approx. 1 … 32 Nm^3/h	System-specific, for detailed information see Section 9.3 Purge air to protect against lens contamination [▶ 41]
Sighting tube connection	G 1¼" inner thread NPT 1¼" female *(option)*	
Purge air connection	G ½" inner thread NPT ½" female *(option)*	

7.4 Integrated flame sensor

Spectral sensitivity range	D–LX 110 UL	185 … 260 nm
	D–LX 110 UA D–LX 710 UA	190 … 520 nm
	D–LX 110 UAF D–LX 710 UAF	280 … 410 nm
	D–LX 110 IS D–LX 710 IS	300 … 1100 nm
	D–LX 110 IG D–LX 710 IG	780 … 1800 nm
Viewing angle	6° 6° horizontal, 12° vertical *(D-LX 110 UL)*	

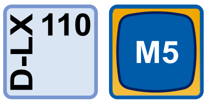
8 Device description

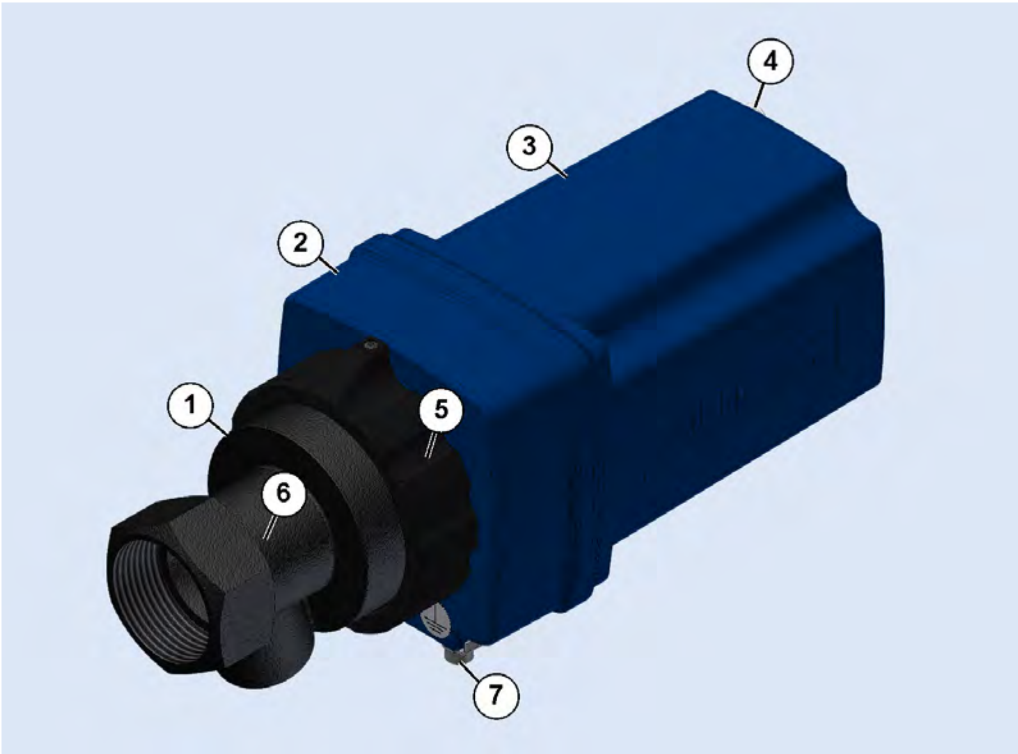
8.1 Scope of supply

The respective scope of supply corresponding to the applicable purchase agreement is listed in the shipping documents included with the delivery. The operating manual is always part of the scope of delivery.

1. Check the delivery for completeness and integrity. Notify the freight forwarder and the DURAG GROUP immediately of any transport damage.

8.2 D-LX 110 with housing M5





1	Purge air flange	5	Connecting nut
2	Middle part	6	Front part
3	Cover	7	Protective conductor connection
4	Plug-in cover (with inspection window)		

Fig. 8.1: The front part (6) and connecting nut (5) together form the purge air flange (1).

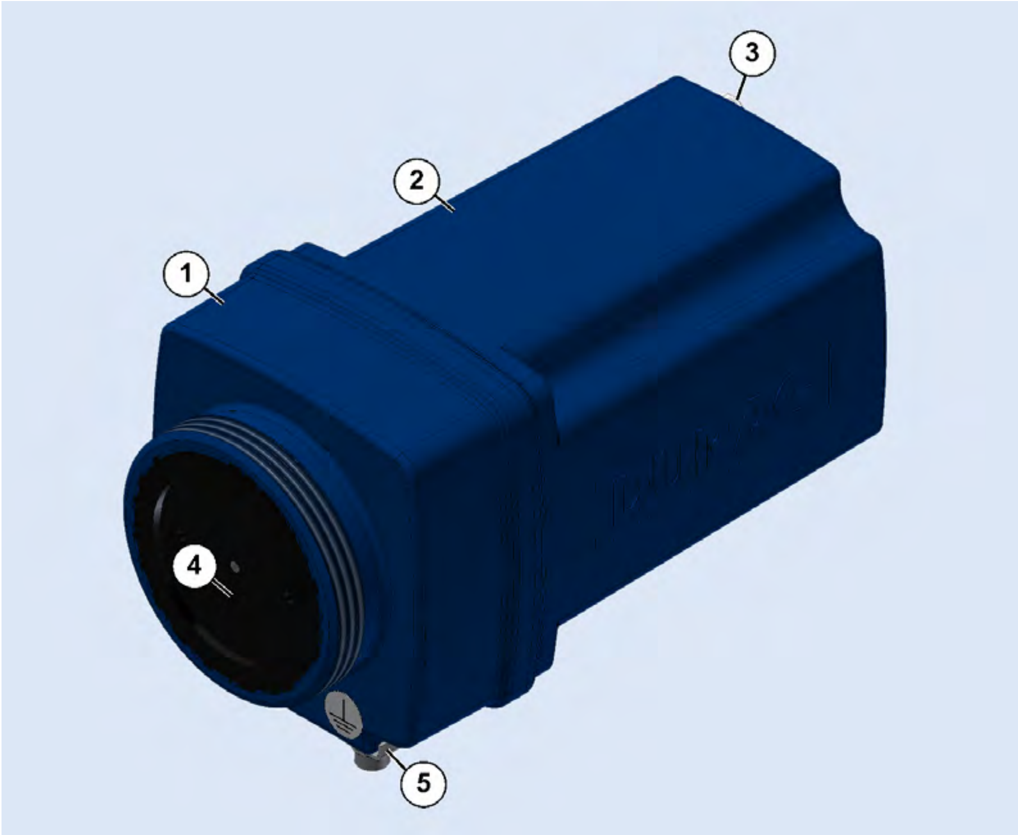
8.3 D-LX 110 with housing M4 for zone 1/21

1	Purge air flange	5	Connecting nut
2	Middle part	6	Front part
3	Ex cover	7	Cable gland
4	Inspection window (not visible)		

Fig. 8.2: The front part (6) and connecting nut (5) together form the purge air flange (1).

8.4 D-LX 710 with housing M5

D-LX 710 M5



1	Middle part	4	Fibre-optic cable adapter
2	Cover	5	Protective conductor connection
3	Plug-in cover (with inspection window)		

Fig. 8.3: Establish a connection to a fibre-optic system using the fibre-optic cable adapter (4) (see Section **8.6 Accessories [▸ 34]**).

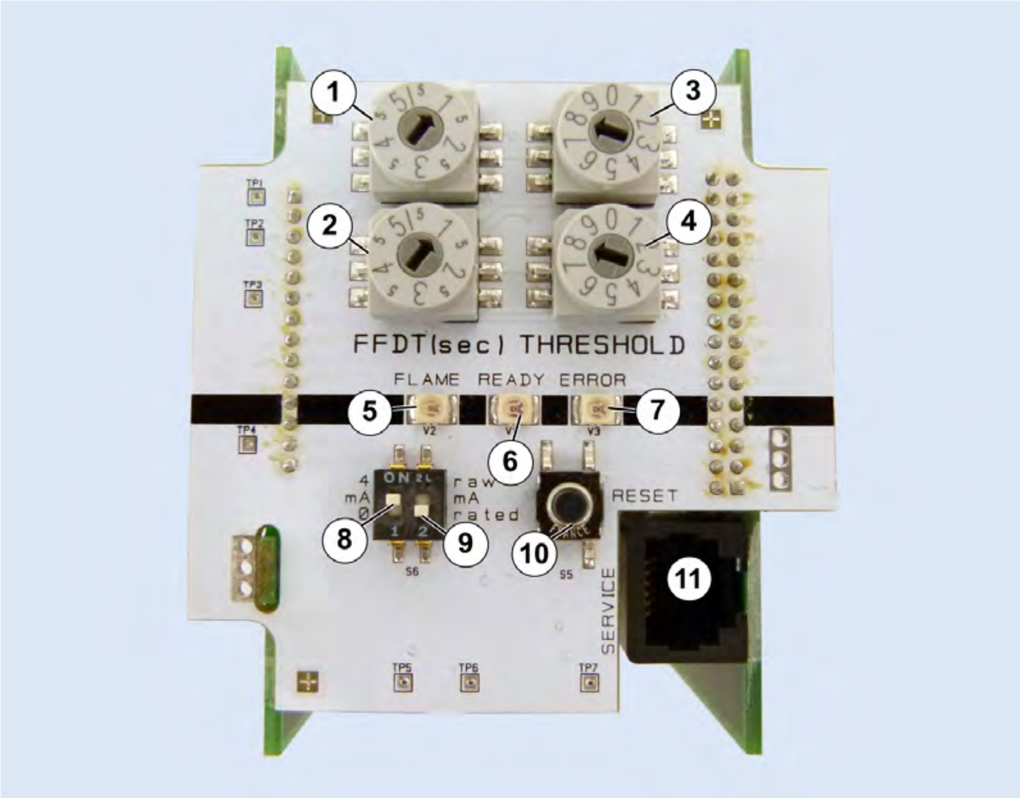
8.5 Front panel (all types)

M5

To operate the switch you must unscrew the plug-in cover (see section 13.1 Removing the plug-in cover (housing M5) [▸ 55]).

M4

To operate the switch you must open the housing (see section 13.5 Opening housing M4 (for zone 1/21) [▸ 58]).

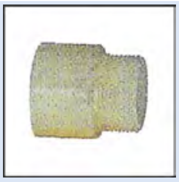


1, 2	Rotary switches; FFDT (see Flame Failure Detection Time) [▸ 101]
3, 4	Rotary switches; threshold
5	LED **FLAME** (green): The flame monitor reports flame ON
6	LED **READY** (yellow): **Permanently illuminated:** The flame monitor is ready for operation **Flashing:** The flame monitor has a fault. Information on the type of fault and rectification measures can be found in Section 17 Fault rectification [▸ 75].
7	LED **ERROR** (red): **Permanently illuminated:** A fault has occurred. The flame monitor has a fault. **Flashing:** The rotary switches have different settings in at least one of the two pairs of rotary switches. This setting difference is causing an error shutdown after 8 s.
8	Switch 0 … 20 mA /4 … 20 mA *(0/4)*
9	Switch for evaluated/raw flame signal *(rated/raw)*
10	Reset
11	Service interface

Fig. 8.4: Front panel with switches and LEDs

8.6 Accessories

D-LX 110	D-LX 710	Item no.	Figure	Name/description
✓		1 101 520		**D-ZS 033-I** Swivel mount with G 1¼" threaded connection
✓		1 115 222	./.	**D-ZS 112** Adapter for fitting a D−LX 110 to a G 1" threaded connection
✓		4 005 322	./.	**D-ZS 113 set** Apertures for inserting in the purge air flange, suitable for DURAG flame monitors, aperture diameters: 2, 3, 4, 5, 7 mm
✓		1 100 517		**D-ZS 133-I** ball valve with G 1¼" threaded connection
	✓	Various		Fibre optic system **D-LL 703**: flexible design, up to 350 °C **D-LL 704**: rigid design, up to 350 °C **D-LL 713 IR**: for space-saving connection on the outside of a burner
✓ UA, UAF, IG, IS	✓ UA, UAF, IG, IS	1 104 669 1 114 903		Combined testing light source for the UV-A, UV-B and IR range for function checking **D-ZS 093:** 230 V~/50 Hz **D-ZS 093:** 115 V~/50 Hz
✓ UL		1107760		Testing light source for the UV-C range for function checking **D-ZS 077-10:** 230 V~/50 Hz **D-ZS 077-10:** 115 V~/50 Hz
✓	✓	4 001 628	./.	**D-NG 24/25a** AC power supply 110 … 240 V~, 50/60 Hz, -40 … 60°C
✓	✓	1 113 984		**D-ZS 140-12** terminal box for flame sensor; IP 65; 12-pole
✓ Ex	✓ Ex	1 114 131		**D-ZS 141-12** Ex connection terminal box for the flame sensor; zone 1, II 2G Ex e IIC T6 Gb, IP65, 12-pin

D-LX 110	D-LX 710	Item no.	Figure	Name/description
✓		1 108 948		D-ZS 117-I heat insulator with potential separation and G 1¼" threaded connection
✓		1 101 525		D-ZS 118 optical adjustment aid for aligning the swivel mount onto the sighting tubes; M72 outer thread
✓	✓	 1 101 528 1 114 329		Light strip indicator for showing the flame intensity (19" system), 3HE/3TE, IP00 D-ZS 129-30: 24V⎓ D-ZS 129-40: 110 … 240 V~
✓	✓	1100506		D-ZS 087-20 Digital display unit for displaying the flame signal
✓	✓	4 017 785	./.	Adapter cable for connecting a D-ZS 087-20 to the D-LX 110/710
✓	✓	1 114 314	./.	Tropicalised varnish for circuit boards

8.7 Options

8.7.1 Connection cables

For device versions with /MP7 at the end of the product code, a connection cable with cable plug will need to be ordered. All other versions are delivered with a connected 2-metre connection cable. Other cable lengths are available:

<table>
<tr><th>D−LX 110</th><th>D−LX 710</th><th>Item no.</th><th>Figure</th><th>Order code</th><th>Length in m</th></tr>
<tr><td colspan="6">Connection cable (PVC, shielded, for cable gland)</td></tr>
<tr><td rowspan="7">✓</td><td rowspan="7">✓</td><td>4 004 739</td><td rowspan="7">./.</td><td>D-LX 110 CBL-V1C-000-03</td><td>3</td></tr>
<tr><td>4 004 960</td><td>D-LX 110 CBL-V1C-000-05</td><td>5</td></tr>
<tr><td>4 004 961</td><td>D-LX 110 CBL-V1C-000-10</td><td>10</td></tr>
<tr><td>4 004 962</td><td>D-LX 110 CBL-V1C-000-15</td><td>15</td></tr>
<tr><td>4 004 963</td><td>D-LX 110 CBL-V1C-000-20</td><td>20</td></tr>
<tr><td>4 004 964</td><td>D-LX 110 CBL-V1C-000-25</td><td>25</td></tr>
<tr><td>4 004 965</td><td>D-LX 110 CBL-V1C-000-30</td><td>30</td></tr>
<tr><td colspan="6">Connection cable for MP7 panel plug (PVC, shielded, with cable bushing)</td></tr>
<tr><td rowspan="8">✓</td><td rowspan="8">✓</td><td>4 004 545</td><td rowspan="8"></td><td>D-LX 110 CBL-V1C-MP7-00</td><td>2</td></tr>
<tr><td>4 004 966</td><td>D-LX 110 CBL-V1C- MP7-03</td><td>3</td></tr>
<tr><td>4 004 967</td><td>D-LX 110 CBL-V1C- MP7-05</td><td>5</td></tr>
<tr><td>4 004 968</td><td>D-LX 110 CBL-V1C- MP7-10</td><td>10</td></tr>
<tr><td>4 004 969</td><td>D-LX 110 CBL-V1C- MP7-15</td><td>15</td></tr>
<tr><td>4 004 970</td><td>D-LX 110 CBL-V1C- MP7-20</td><td>20</td></tr>
<tr><td>4 004 971</td><td>D-LX 110 CBL-V1C- MP7-25</td><td>25</td></tr>
<tr><td>4 004 972</td><td>D-LX 110 CBL-V1C- MP7-30</td><td>30</td></tr>
</table>

8.7.2 Factory settings

	Possible adjustment values	Factory settings
Analogue output (rated) in mA	0 … 20; 4 … 20	4 … 20
Flame Failure Detection Time (FFDT) in s	0.5; 1; 1.5 … 5	1
Threshold for flame intensity	0, 1, 2 … 9	8

If no other specifications are entered with the order, the flame monitor will be supplied with the factory settings.

Devices with

- **FM approval** (regional code: H): Select max. 4 s for FFDT!
- **AGA approval** (regional code: C) and for simultaneous use in Australia: Select max. 2 s for FFDT!

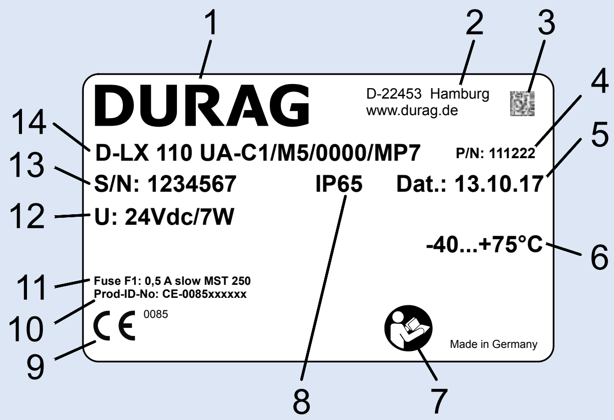
Information on the regional code can be found in Section 8.8 Type code [▸ 37].

8.8 Type code

The type designations take the form of D-LX bbb ccc-de/ff/gggg/hhh.

Position	Entry	Meaning		
bbb	Device variant			
	110	For direct optical view with a lens		
	710	For optical access via a fibre optic system		
ccc	Spectral sensitivity range in nm			
	UA	190 … 520		
	UAF	280 … 410		
	UL	185 … 260 *(D-LX 110 only)*		
	IG	780 … 1800		
	IS	300 … 1100		
-d	Regional code: certifications necessary for the region have been obtained			
	C	EU and Australia	L	EAU
	H	USA and Canada	T	India
	K	Korea		
e	Information on the presence of certificates or external confirmations of functional safety (SIL)			
	1	Present		
/ff	Housing characteristics (material, shape)			
	M4	Aluminium housing		
	M5	Aluminium housing with plug-in cover (only with /gggg=0000 or 87Ex or 86Ex)		
	JZ	Special version for customer		
/gggg	Suitability for specific ambient conditions			
	0000	No explosion protection		
	87Ex	Gas and dust explosion protection IECEx and II 3 GD in accordance with ATEX		
	86Ex	Gas, dust and fibre explosion protection Div. 2 in accordance with NEC 500		
	85Ex	Gas, dust and fibre explosion protection Div. 1 in accordance with NEC 500		
	84Ex	Gas and dust explosion protection IECEx and II 2 GD in accordance with ATEX		
/hhh	Code for the signal and supply connection			
	NPT	NPT thread (adapter) with transport protection for M5 housing		
	M20	M20 thread with transport protection		
	PCG	Plastic cable gland		
	MCG	Metal cable gland		
	MP7	12-pin metal panel plug, up to 50 V		

8.9 Type label



1	Manufacturer	8	IP code
2	Manufacturer address	9	CE mark, examining body (no.)
3	2D code (for internal purposes)	10	Product identification number (DVGW)
4	Item number	11	Device fuse
5	Date of manufacture	12	Operating voltage/max. power consumption
6	Ambient temperature	13	Serial number
7	Before starting any work, read the operating manual!	14	Device type

Fig. 8.5: Type label (example)

Information on the contents of the Ex type label can be found in section 4.8 Ex type label [▸ 23].

9 Functional description

The compact flame monitor is a safety device consisting of a control unit and an integrated optical flame sensor located inside a housing. This integration saves time and costs for assembly and wiring compared to the setup with separate flame sensor and control unit. The flame sensor converts the light signal from the flame into an electrical signal, while the control unit ensures self-monitoring, evaluates the Flame signal [▸ 101] and communicates the results.

D-LX 110

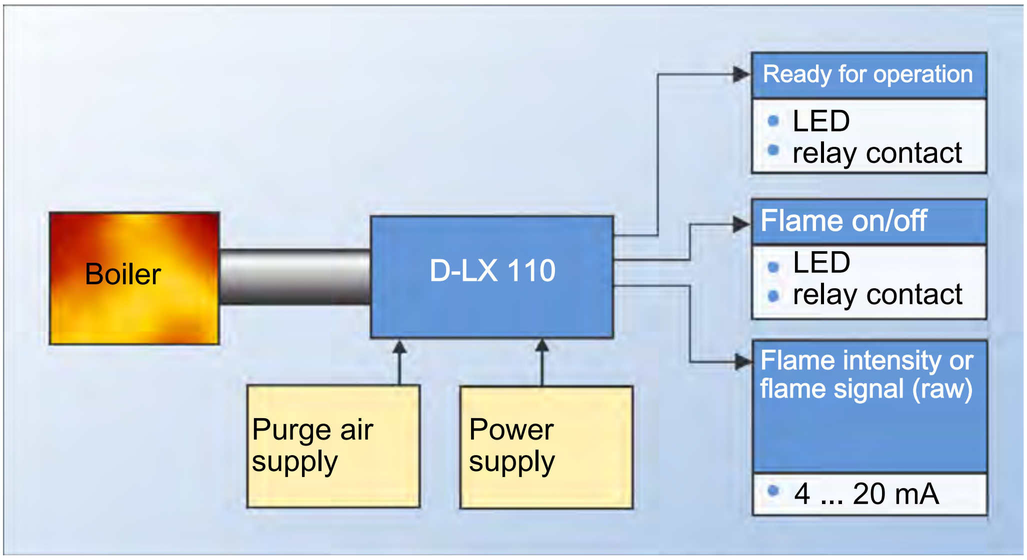


Fig. 9.1: D-LX 110 fitted on the boiler (schematic depiction)

D-LX 710

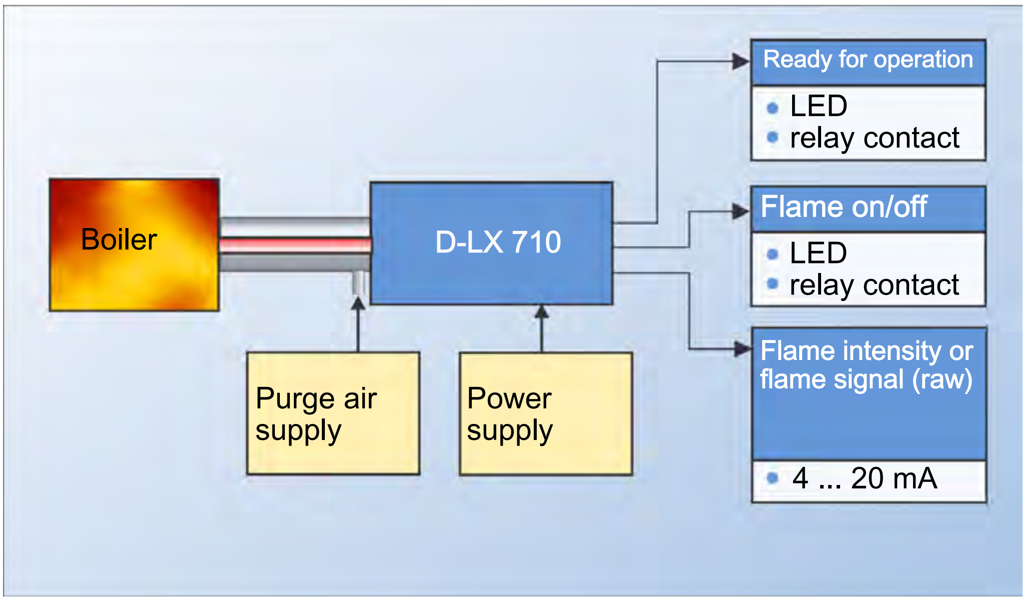


Fig. 9.2: D-LX 710 fitted on the boiler (schematic depiction)

Different spectral ranges of the flame (UV, visible light or IR) can be detected depending on the flame sensor fitted in the flame monitor. The question of whether the static or only the dynamic portion of the Flame radiation [▸ 101] is detected also depends on the flame sensor.

Evaluation of the flame signal

The flame monitor evaluates the flame signal using a **switching threshold**. This defines whether the flame signal should lead to a *flame ON* or *flame OFF signal*. Only flame signals that are higher than the switching threshold can be evaluated as *Flame ON*.

The **Flame Failure Detection Time (FFDT)** (formerly: Safety time [▶ 101]) is the time it takes the flame signal to respond to, for example, a flame failure or an impermissible impairment of the combustion process. As soon as the flame signal falls below the switching threshold, the FFDT [▶ 101] starts to count down. If the flame signal remains below the switching threshold during the FFDT, the flame monitor will output the *Flame OFF* signal once the FFDT has elapsed.

Different values can be chosen for the switching threshold and the FFDT. This allows the user to select a suitable combination of FFDT and switching threshold for the plant.

Output of the flame signal and its evaluation

The result of the flame evaluation (*Flame ON* or *Flame OFF*) is provided as

- Positively driven relay contact
- Green illuminated **FLAME** display on the front panel

The analogue output (4... 20 mA or 0…20 mA) can deliver either:

- The evaluated flame signal as a function of the switching threshold (flame intensity)
- The raw flame signal (pulse rate of the flame sensor)

Self-monitoring

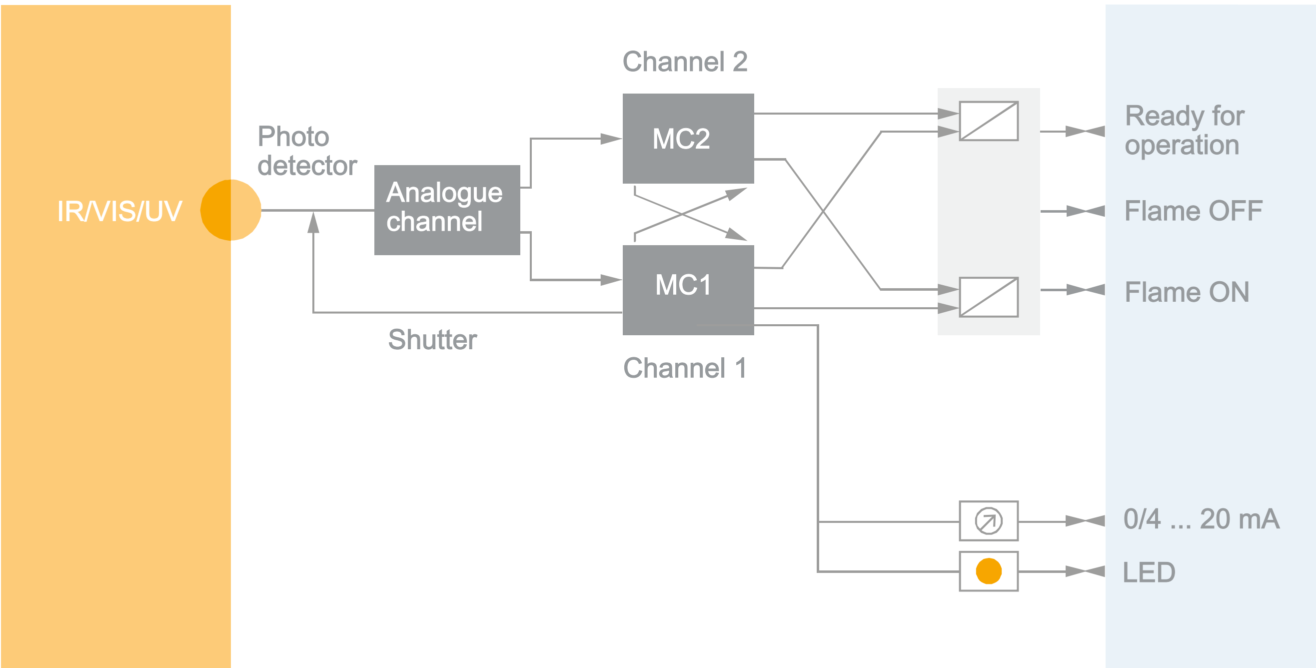


Fig. 9.3: Functional diagram

In order to ensure a fail-safe design, the flame monitor has a dual-channel construction. The two microprocessor systems control all functions and self tests, and monitor all safety-relevant time sequences. The input and output states are checked by the two microcontrollers independently from one another and are then compared. Operation is only continued if the states match.

The flame monitor can bridge failures in supply voltage of up to 30 ms.

9.1 Ultraviolet flame sensor

A flame's UV radiation zone is generally considerably smaller than its IR zone. Fixtures and boiler walls also do not emit any UV radiation. UV flame monitors are therefore very selective, i.e. insensitive to background radiation [▶ 101].

In D-LX 110 UL, a phototube with a spectral sensitivity of λ= 185 to 260 nm is used. Because this spectral range is so narrow and so far away from daylight, it is permissible to evaluate both the dynamic and the static flame radiation. This makes the D-LX 110 UL a highly sensitive flame monitor with excellent selectivity for all UV-C-emitting flames such as gas and oil.

In **D-LX 110 UA** and **D-LX 710 UA** a photodiode with a spectral sensitivity of λ= 190 to 520 nm is used. The integrated flame sensor thereby records the dynamic blue to transparent radiation range, e.g. of gas, oil and coal flames, without suffering any major signal failures as a result of water vapour, recirculation gas or similar UV-absorbing gases.

In **D-LX 110 UAF** and **D-LX 710 UAF**, the D-LX 110 UA and D-LX 710 UA photodiode with optical filter is used. This filtration brings about a limited spectral range of λ= 280 to 410 nm, which gives better results with very intensive UV radiation or with higher selectivity requirements.

> The filter is impermeable in the visible spectral range, and therefore appears black to the human eye. The photodiode is visible through the flame monitor lens.

9.2 Infra-red flame sensor

A flame's IR radiation zone is often extensive, and is very intensive when compared to UV radiation. The IR zone is therefore easy to detect from various different viewing angles, has a strong signalling effect and is insensitive to absorption by gases. The IR flame sensor, however, is more sensitive to background radiation when compared to a UV flame sensor.

If a flame appears yellow or red, then it is "visible" for this type of flame monitor. However, this requires movement (dynamic) within the flame.

In D-LX 110 IS and D-LX 710 IS, a photodiode with a spectral sensitivity of λ= 300 to 1100 nm is used. The integrated flame sensor therefore detects the radiation range that is visible to the human eye.

In D-LX 110 IG and D-LX 710 IG, a photodiode with a spectral sensitivity of λ= 780 to 1800 nm is used. The integrated flame sensor can therefore detect the dynamic radiation range generated by almost all fuels. This flame monitor has a strong signalling effect but lower selectivity due to the extensive IR emission zones.

Flames that involve the absorption of short-wave UV radiation by dust, water vapour or other substances can be monitored in the IR range. Application examples include waste incineration plants and wood combustions.

With NOx-reduction measures for the combustion, such as exhaust gas recirculation or systems with combination burners for gas and oil, the IR flame sensors in particular have proved themselves suitable for flame monitoring with a spectral sensitivity of up to 1800 nm.

9.3 Purge air to protect against lens contamination

D-LX 110

To ensure undisturbed operation, the flame monitor's sighting tube and lens must be protected against contamination from the combustion chamber. This is ensured by the purge air. It must flow from the inlet into the purge air flange towards the combustion chamber such that it can also remove heavy particles. For combustions with low particle shares (e.g. gas firings), sufficient purging has been achieved with flow velocities of v=1 m/s in a 1¼" sighting tube (air consumption approx. 3 m^3/h). Firing systems with a higher particle load, such as coal firing systems, require larger amounts of purge air (flow velocity v = 3 m/s). The flow velocity information for the purge air in the sighting tube are merely reference values here. The flow velocities must always be adapted to the plant conditions.

Purge air with an appropriate upstream pressure is required in order to ensure the desired air flow through the sighting tube (see above for examples). This also depends on the pressure in the combustion chamber. The differential pressure between the purge air inlet at the purge air connection and the outlet from the sighting tube into the combustion chamber is decisive. You can find the required differential pressure in the following diagram:

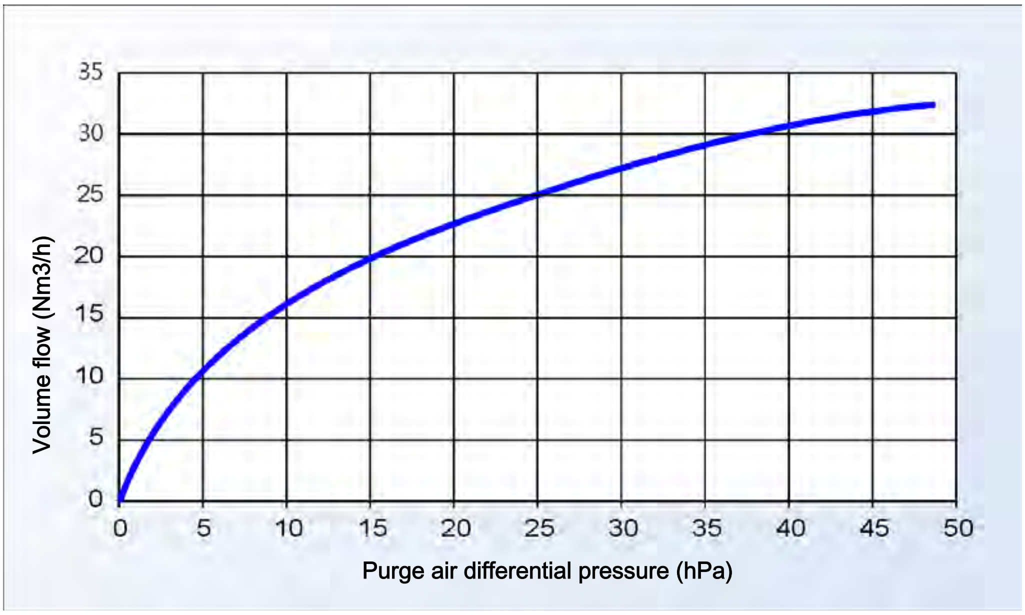


Fig. 9.4: Purge air flow through the sighting tube depending on the differential pressure

The numerical values relate to a sighting tube of 1 ¼“ in diameter and 500 mm in length. If using sighting tubes with the same diameter and between 300 and 800 mm in length, the change to the volume flow would be negligible, especially with low differential pressures.

10 Transport, packaging and storage

1. Take care when transporting the device. Avoid rough impacts.

Transport inspection

The scope of delivery is specified on the delivery note. The operating manual is always part of the scope of delivery.

1. Check the delivery for completeness and integrity. If possible, photos should be used to document any damage to the goods or packaging.

Visible transport damage:

1. Immediately submit a complaint to the forwarding agent and DURAG GmbH.
2. Only accept the delivery conditionally.
3. Note the extent of the damage on the transport documents or on the carrier's delivery note.
4. Initiate the complaint.

Concealed transport damage:

1. Submit a complaint within 7 days.

Packaging

The individual packages are packaged according to the anticipated transport conditions. The packaging should protect the individual components against transport damage and corrosion until installation.

1. Therefore, do not destroy the packaging. Only remove it shortly before assembly.
2. If possible, also use the original packaging for future transport. The materials and the moulded parts used for some packages ensure safe transport.

Handling packaging materials

Packaging materials are valuable raw materials and can be recycled.

1. Dispose of the packaging materials in an environmentally friendly manner, observing the applicable local disposal regulations.

Storage conditions for the device and spare parts

- Do not store outdoors.
- Store in a dry, dust-free location.
- Do not expose to aggressive media.
- Do not allow the temperature to fall below the dew point.
- Protect against mechanical damage.
- Storage temperature: as per ambient temperature for operation
- Relative humidity: max. 60%, non-condensing
- For storage periods of over 3 months: Regularly check the condition of all parts as well as the packaging. The transport packaging is not normally suitable for long-term storage.

Maximum storage time (D-LX 110 UL*** only)

Store the flame monitor for a maximum of four years: The UV cell ages even if not being used!

11 Installation

Select a suitable installation site

The specification of the right viewing position is part of ensuring optimal and selective flame monitoring. Make sure that the flame is easy to detect for the flame monitor in all of the burner's load ranges and that there are no flames from neighbouring burners in its monitoring axis where possible.

The flame monitor has a viewing angle of approx. 6°. This must always be aligned to the area at the base of the flame (the flame's ignition point), i.e. the first third of the flame. Information on the dimensions can be found in chapter 20 Dimensional diagrams [▶ 80].

D-LX 110

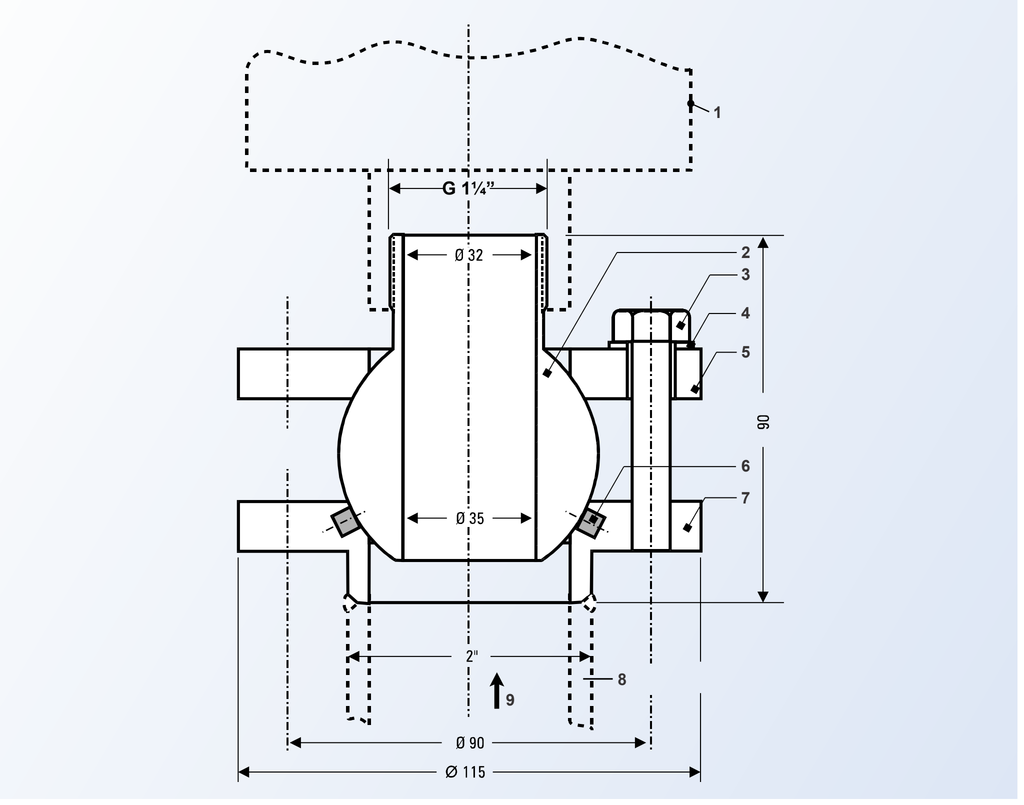
Determine an appropriate upstream pressure for the purge air. Assistance is provided in chapter 9.3 Purge air to protect against lens contamination [▶ 41].

11.1 Useful accessories

D-LX 110

Optimally aligning the flame monitor to the flame

Different swivel mounts in series D-ZS 033 are available (see section 8.6 Accessories [▶ 34]), e.g. swivel mounts D-ZS 033-I:



1	Flame monitor	6	Sealing cord
2	Ball joint	7	Welded flange
3	Hexagon screw (3x),tightened to maximum 20 Nm	8	Sighting tube
4	Shim (3x)	9	Flame
5	Flange		

Fig. 11.1: Swivel mount D−ZS 033−I in installed condition

Component	Material (properties)
Welded flange	1.0570, galvanised surface
Flange	Al, alloy 230
Ball joint	Al, alloy 230
Hexagon screw	M10x60, DIN 558
Shim	10.5, DIN 125
Sealing cord	Graphite band, asbestos-free

Table 11.1: Component specifications

Use at high temperatures

If flame monitors need to be mounted onto sighting tubes with high operating temperatures, then the purge air also acts as a cooling mechanism. In these cases, it is recommended that heat insulator D-ZS 117-I (see section 8.6 Accessories [▸ 34]) be screwed in between the flame monitor and the sighting tube or swivel mount D-ZS 033-I. Always make sure that the maximum operating temperature of the flame monitor (see Section 7 Technical data [▸ 27]) is not exceeded.

Compensating currents on the earth conductor generate electromagnetic disturbances

Electromagnetic disturbances (e.g. caused by currents from an ignition device against protective conductor potential) can impair the function of **flame monitors with metal housings (housing M4 and M5)**. The electromagnetic disturbances can spread to the flame monitor via the purge air flange. The D-ZS 117-I isolator can be installed between the sighting tube and the flame monitor in order to prevent this. A purge air line made from non-conductive materials should also be used.

Overpressure in the combustion chamber

If an overpressure occurs at the flame monitor sighting tube, it is recommended that ball valve D-ZS 133-I (see section 8.6 Accessories [▸ 34]) be installed between the flame monitor and the sighting tube or swivel mount D-ZS 033-I. If the lens becomes contaminated, it can be safely cleaned once the ball valve has been closed.

11.2 Fitting the flame monitor onto the burner

Make the necessary preparations based on the local conditions, the design of the combustion chamber and the devices used to enable the flame monitor to be installed in the combustion chamber. In particular, all adjacent components must be cooled.

Information on the installation of the D-LX 710 can be found in the operating manual for the fibre optic system D-LL 703/704.

D-LX 110

The flame monitor is installed onto a sighting tube with a G1¼" outer tube thread.

Required: 2-mm socket head wrench

1	Front part thread	4	Connecting nut
2	Front part	5	Locking screw
3	Purge air connection	6	Middle part

Fig. 11.2: Middle part screwed onto the front part

1. Tightly screw the front part (**2**) onto the front part thread (**1**) on the sighting tube.
2. Screw the middle part (**6**) onto the front part by hand with the connecting nut (**4**).
3. In order to prevent the middle part becoming loose unintentionally, the connecting nut is fitted with a locking screw (**5**). Tighten the locking screw.

11.3 Connect the purge air line

D-LX 110

For undisturbed operation, you need purge air with adequate upstream pressure. (see chapter 9.3 Purge air to protect against lens contamination [▶ 41]).

NOTICE

The system may be damaged and contaminated due to failure of the purge air!

If the purge air fails, the lens may quickly become contaminated and the flame monitor may overheat due to a build-up of heat and be destroyed. Therefore:
Make sure that

- the purge air supply is operating reliably and smoothly (even when the system is stopped).
- you immediately detect a failure of the purge air.
- the flame monitor is protected by a valve or dismantling process in the event of a purge air failure.

Fig. 11.3: Purge air connection

1. Connect the flame monitor to the purge air supply. To do so, attach a hose from the plant's purge air blower onto the purge air connection (1) (½" inner thread).

D-LX 710

When using a fibre-optic system, the purge air supply for the D−LX 710 is guaranteed by the corresponding fibre-optic-cable supply. Installation information can be found in the description of the optional fibre-optic system D-LL 703/704.

11.4 Establishing electrical connections

The contact outputs 'flame OFF', 'flame ON' and 'ready for operation' are protected against excessive electrical currents by the fuse F3.

1. Select a fuse F3 that is suitable for your application (see Section 7 Technical data [▶ 27]). Information on replacing the fuse can be found in Section 17.1 Replacing the electrical fuse F1 [▶ 76].

11.4.1 Safety

The electrical installation of the flame monitor must be performed by a specialised electrician (see Section 2.4 Qualified personnel [▶ 11]) in accordance with local regulations.

The electrical installation of the flame monitor in hazardous areas must be performed by a specialised electrician with additional knowledge of explosion protection (see Section 2.4 Qualified personnel [▸ 11]) in accordance with local regulations.

DANGER

Risk to life due to electric current!

There is an immediate risk to life when touching parts carrying live voltage. Damage to the insulation or individual components can present a risk to life. Therefore:

- Before connecting the flame monitor, de-energise the electrical system, check to verify the voltage-free state, and secure against re-activation. **As well as the supply voltage for the flame monitor, the supply voltage for the contact circuit must also be deactivated without fail!**

It is essential to observe the following:

- Direct switching of fuel valves is not permitted.
- The flame signal is safety-relevant. A safety-relevant evaluation is only allowed via the relay contact for the flame ON signal. The contact for the flame OFF signal and the analogue output are not suitable here! They are solely for information purposes!
- The operating voltage must meet the requirements of a protective extra-low voltage (PELV) system (in accordance with IEC 60730-1)! This is the only way of safely making setting adjustments on the flame monitor during operation when the plug-in cover is removed.
- The flame monitor is constructed with potential separation between the supply voltage (24 V) and the device's internal voltage. The analogue output has an electrically conductive connection to the device's internal voltage. Therefore, do not connect the blue and grey wires (terminal X1.2 and X1.3, see Section 19 Circuit diagrams [▸ 78]) with one another. This may damage the flame monitor!
- Always connect the functional earth and shielding to earth.
- Connect the external earth conductor separately on the outside of the housing.
  Cross-section:
  4 mm^2: with unprotected cables
  1 mm^2: with sheathed cables
  Use a suitable terminal fitting for copper earth connectors. Connect the device's internal functional earth in the sub-distribution unit to earth. The external earthing is also required in order to ensure electromagnetic interference immunity!
- Route the cables so that high electrical currents and high data rates in other lines cannot have an impact.
- Route the cables so as to exclude the risk of accidents as a result of tripping on them or becoming entangled in them.

11.4.2 Cables and cable glands for devices without explosion protection

The flame monitor is delivered with one of the following options:

- Already wired with a connection cable
- With a panel plug (model ***/MP7) with separate cable and pre-assembled cable jack
- Without connection cable and without cable gland

Device is completely wired with connection cable

1. Establish the electrical connections to the plant in accordance with the circuit diagram (see sticker on the housing or Section 19 Circuit diagrams [▸ 78]).

✓ The flame monitor is now ready for initial commissioning.

Device with panel plug (*Model* ***/MP7)

1. Establish the electrical connections to the plant in accordance with the circuit diagram (see sticker on the housing or Section 19 Circuit diagrams [▸ 78]).
2. Insert, shake and screw in the panel plug repeatedly in order to ensure that it is fully inserted into the cable jack. This is the only way of achieving the required IP protection!

✓ The flame monitor is now ready for initial commissioning.

Device without connection cable and without cable gland

1. Remove the cover (see Section 13.3 Removing the cover (housing M5) [▸ 56].

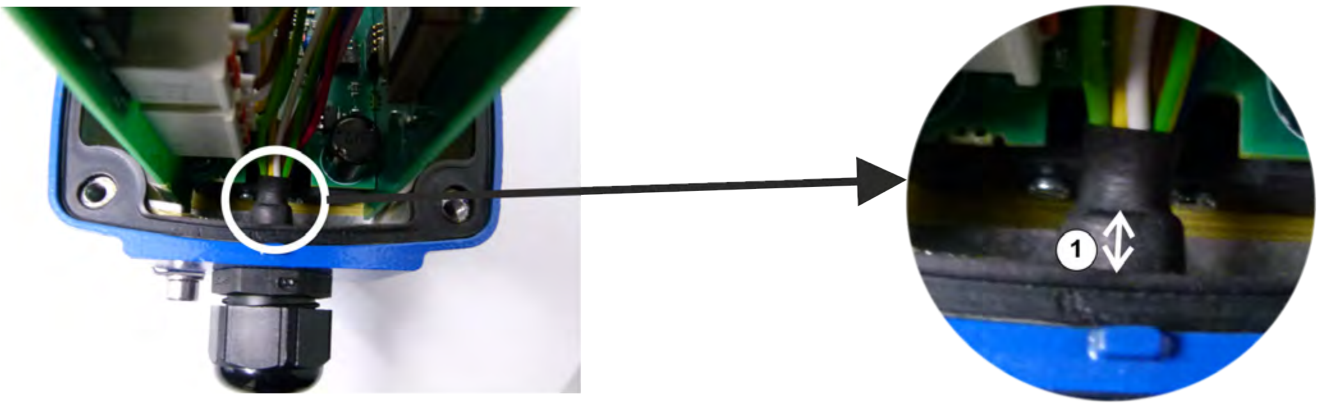


Fig. 11.4: A flame monitor with completely wired cable is shown. For models ***/M20 cable gland and cable are not included in the scope of delivery!

- Make sure, that the distance (**1**) between the end of the cable jacket and the metal housing at least reaches three mm. Only so the increased electrical insulation by distance of the wires to the metal housing is obtained!
- Use your own cable gland respectively your own pipe system
  - with M20x1.5 for model ***/M20
  - with NPT ½“ for model ***/NPT
- Use a proper cable (see Section 11.4.5 Requirements for fixed cables [▸ 53]).
- Make sure, that the wished degree of protection is reached (see section 7.1 Devices with housing M5 [▸ 27]).
- **Dismantling of the cable:** Meet the following lengths of the wires:
  - Wires green, yellow, white, brown (clamps X6.1 – X6.4): **max. 9 cm** (due to electrical safety)
  - All other wires: **max. 12.5 cm** (due to EMC stability)

1. Establish the electrical connections to the plant in accordance with the circuit diagram (see sticker on the housing or Section 19 Circuit diagrams [▸ 78]).
2. Screw down the cover (see Section 13.4 Screwing down the cover (housing M5) [▸ 57]).

✓ The flame monitor is now ready for initial commissioning.

11.4.3 Cables and cable glands for zone 2/22 and Div. 2

You will normally use the flame monitor with the factory-supplied cable and the cable gland for zone 2/22.

1. Securely route the cable. Ensure suitable strain relief.
2. You can find information on the ambient temperature of the cable in 11.4.5 Requirements for fixed cables [▸ 53].

Alternatively (and when using D-LX*/86Ex/* for Div. 2) you can also use in-house components:

WARNING

Risk of explosion due to unsuitable cables and cable glands!

Cables and cable glands must not counteract the special properties of the housing's ignition protection type: Therefore:

- Only use suitable components.
- Install the components correctly. (See also Section 4.6 Ignition protection categories [▸ 17]).

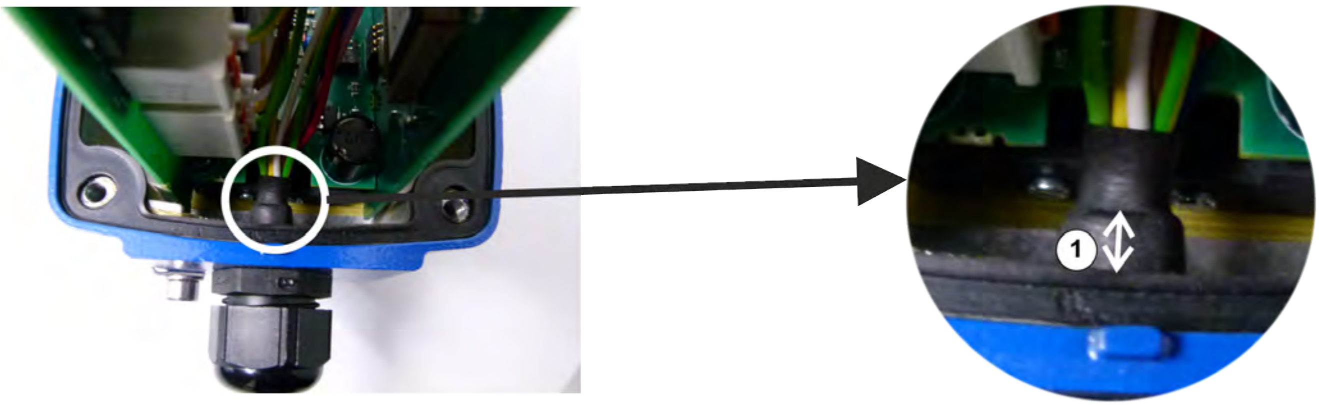


Fig. 11.5: A flame monitor with completely wired cable is shown. For models ***/M20 cable gland and cable are not included in the scope of delivery!

1. Make sure, that the distance (**1**) between the end of the cable jacket and the metal housing at least reaches three mm. Only so the increased electrical insulation by distance of the wires to the metal housing is obtained!
2. Remove the cover (see Section 13.3 Removing the cover (housing M5) [▸ 56]).

Using an in-house cable and the supplied cable gland

- Use a suitable cable (see Section 11.4.5 Requirements for fixed cables [▸ 53]). Take special care to comply with the permissible cable diameter for the cable gland.
- **Dismantling of the cable:** Meet the following lengths of the wires:
  - Wires green, yellow, white, brown (clamps X6.1 – X6.4): **max. 9 cm** (due to electrical safety)
  - All other wires: **max. 12.5 cm** (due to EMC stability)
- Use the seal supplied with the cable gland. Explosion protection is only guaranteed together with the seal.

Using an in-house cable gland and an in-house cable

- Use a cable gland or pipe system with thread M20x1.5 (gland size 20).
- Use a suitable cable (see Section 11.4.5 Requirements for fixed cables [▸ 53]).
- **Dismantling of the cable:** Meet the following lengths of the wires:
  - Wires green, yellow, white, brown (clamps X6.1 – X6.4): **max. 9 cm** (due to electrical safety)
  - All other wires: **max. 12.5 cm** (due to EMC stability)
- A separate test certificate must be provided for the cable gland. The cables and cable glands must be selected so as not to change the protection class of the device.
- For maximum ambient temperatures Ta_{max} > 60 °C:
  - Use cables with $T_{max} = Ta_{max} + 15$ K
  - Use a cable gland with $T_{max} = Ta_{max} + 10$ K

    Example: For operation of the device up to Ta_{max} = 70 °C the cable must be approved up to 85 °C and the cable gland up to 80 °C.

D-LX*/86Ex/* for Div 2

- The flame monitor is fitted with an adapter for a 1/2" thread. Make sure to use a suitable cable and cable gland so as not to change the protection class of the housing.

All devices

1. Establish the electrical connections to the plant in accordance with the circuit diagram (see sticker on the housing or Section 19 Circuit diagrams [▶ 78]).
2. Screw down the cover (see Section 13.4 Screwing down the cover (housing M5) [▶ 57]).

✓ The flame monitor is now ready for initial commissioning.

11.4.4 Cables and cable glands for zone 1/21 and Div. 1 (housing M4)

You will normally use the flame monitor with the factory-supplied cable and the cable gland for zone 1/21. Observe the following:

1. You can find information on the ambient temperature of the cable in Section 11.4.5 Requirements for fixed cables [▶ 53].
2. The cable must have a minimum length of 0.5 m.

Alternatively (and when using D-LX*/85Ex/* for Div. 1) you can also use in-house components:

WARNING

Risk of explosion due to unsuitable cables and cable glands!

Cables and cable glands must not counteract the special properties of the housing's ignition protection type: Therefore:

- ▶ Only use suitable components.
- ▶ Install the components correctly. (See also Section 4.6 Ignition protection categories [▶ 17]).

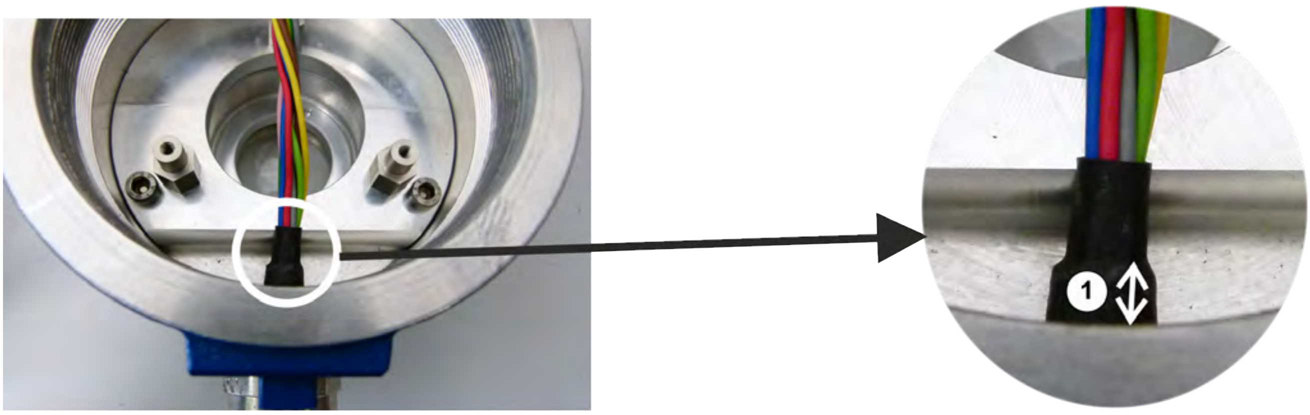


Fig. 11.6: A flame monitor with completely wired cable is shown. For models ***/M20 cable gland and cable are not included in the scope of delivery!

1. Make sure, that the distance (1) between the end of the cable jacket and the metal housing at least reaches three mm. Only so the increased electrical insulation by distance of the wires to the metal housing is obtained!
2. Open the housing (see chapter 13.5 Opening housing M4 (for zone 1/21) [▶ 58]).

Using an in-house cable and the supplied cable gland

- Use a suitable cable (see Section 11.4.5 Requirements for fixed cables [▸ 53]).
Take special care to comply with the permissible cable diameter for the cable gland.
- **Dismantling of the cable:** Meet the following lengths of the wires:
  - Wires green, yellow, white, brown (clamps X6.1 – X6.4): **max. 9 cm** (due to electrical safety)
  - All other wires: **max. 12.5 cm** (due to EMC stability)
- Use the seal supplied with the cable gland. Explosion protection is only guaranteed together with the seal.

Using an in-house cable gland and an in-house cable

Observe the following:

- The used cable gland must meet the requirements of EN 60079-1 Sections 13.1 and 13.2. A separate test certificate must be provided.
- Use a cable gland or pipe system with thread M20x1.5 (gland size 20). Threads, e.g. for cable glands in the Ex-d area, form gaps. At least five fully functioning meshed threads are required in order to ensure sufficient safety in the Ex d area.
- Use a suitable cable (see Section 11.4.5 Requirements for fixed cables [▸ 53]). Take special care to comply with the permissible cable diameter for the cable gland.
- **Dismantling of the cable:** Meet the following lengths of the wires:
  - Wires green, yellow, white, brown (clamps X6.1 – X6.4): **max. 9 cm** (due to electrical safety)
  - All other wires: **max. 12.5 cm** (due to EMC stability)
- The cables and cable glands must be selected so as not to change the protection class of the device.
- For maximum ambient temperatures Ta_{max} > 60 °C:
  - Use cables with $T_{max} = Ta_{max} + 15$ K
  - Use a cable gland with $T_{max} = Ta_{max} + 10$ K
    Example: For operation of the device up to Ta_{max} = 70 °C the cable must be approved up to 85 °C and the cable gland up to 80 °C.

D-LX*/85Ex/* for Div 1

- The flame monitor is fitted with an adapter for a 1/2" thread. Make sure to use a suitable cable and cable gland so as not to change the protection class of the housing.

All devices

1. Establish the electrical connections to the plant in accordance with the circuit diagram (see sticker on the housing or Section 19 Circuit diagrams [▸ 78]).
2. Close the housing (see Section 13.6 Closing housing M4 (for zone 1/21) [▸ 60]).

✓ The flame monitor is now ready for initial commissioning.

11.4.5 Requirements for fixed cables

DURAG offers special cables for the flame monitor. If there are reasons why these cannot be used, alternative cables must meet at least the following requirements:

Intended cable		Sheathed cable	Sheathed cable with max. 50 V or PELV on contact circuit*	Individual conductors in a conduit	Individual conductors in a conduit with max. 50 V or PELV on contact circuit*
For device model		Device without NPT connection		Device with NPT connection (D-LX*/NPT)	
Minimum cross-section in mm², stranded wire with wire end ferrule					
X1.1	Red wire	0.5 … 0.75			
X1.2	Blue wire	0.5 … 0.75			
X1.3	Grey wire	0.5 … 0.75			
X1.4	Pink wire	0.5 … 0.75			
X6.1*	**Green wire**	0.75 … 1.5	0.5 … 0.75	0.75 … 1.5	0.5 … 0.75
X6.2*	**Yellow wire**				
X6.3*	**White wire**				
X6.4*	**Brown wire**				
X4.1/X4.2	Green/yellow wire		0.75		0.75
Connecting terminals: Cable cross-section from - to/with/without plastic collar on the wire end ferrule; Stripping lengths					
Wires x1.1-x1.4		0.25 … 0.75/1.5 mm²/AWG 24-14; stripping length: 8 … 9 mm			
Wires x6.1-x4.2		0.25 … 1.5/2.5 mm²/AWG 24-12; stripping length: 9 … 10 mm			
Structure					
Connection to the housing		M20x1.5 or NPT 1/2"		Conduit NPT 1/2"	
Sheathed cable		Yes		No	
Sheath diameter in mm, suitable for cable gland		7 … 13		-	-
Screen		Yes		No	
Stranding in pairs		Yes		(yes)	(yes)
Max. contact circuit voltage (wires x6.1-x6.4) in V		250	50	250	50
Insulation/electric strength between the wires in V (at 50 Hz, 60 s)					
Wire - wire, only x1.1-x1.4		min. 1000			
Wire - wire, only x6.1-x6.4		min. 2000	min. 1500	min. 200	min. 1500
Wire - wire, (x1.1-x1.4) to (x6.1-x6.4), (x1.1-x1.4) to 9 and (x6.1-x6.4) to 9		min. 2900	min. 2500	min. 2900	min. 2500
Wire – shielding		min. 1450	min. 1250	min. 1450	min. 1250
Temperature range in °C		-40 … +85 with fixed routing			

* Wires x6.1 to x6.4 are part of the contact circuit.

12 Initial commissioning

DURAG can perform the initial commissioning upon request. The on-site electrical installation must, if necessary, be performed by qualified electrical companies and must comply with the local regulations.

1. Make sure that the flame monitor is installed as described in chapter 11 Installation [▸ 44].
2. Supply the flame monitor with purge air. You can find information on the operating pressure and consumption in chapter 7.3 Purge air and pipe connections [▸ 29].
3. Switch on the operating voltage (see type label or Section 7 Technical data [▸ 27]). The operating voltage must meet the requirements of a protective extra-low voltage (PELV) system! This is the only way of safely making setting adjustments on the flame monitor during operation when the plug-in cover is removed.

✓ The flame monitor is ready for operation immediately. In many cases, the factory setting is suitable for operation.

WARNING

Wrong evaluation of the flame signal due to incorrectly set flame monitor!

An incorrectly set flame monitor will provide incorrect indications of the burner states.

Therefore:

- Make sure that the flame monitor reliably detects the burner states of flame ON/OFF **in all operating states**. Information on modifying the flame monitor to meet plant-specific conditions can be found in Section 14 Setup [▸ 62].
- Secure the flame monitor against unauthorised modifications being made to its alignment to the flame and to the selected settings on the device.

13 Opening and closing the housing

13.1 Removing the plug-in cover (housing M5)

M5

Any work carried out when the plug-in cover is open must be performed by an electrically trained person (see section 2.4 Qualified personnel [▶ 11]).

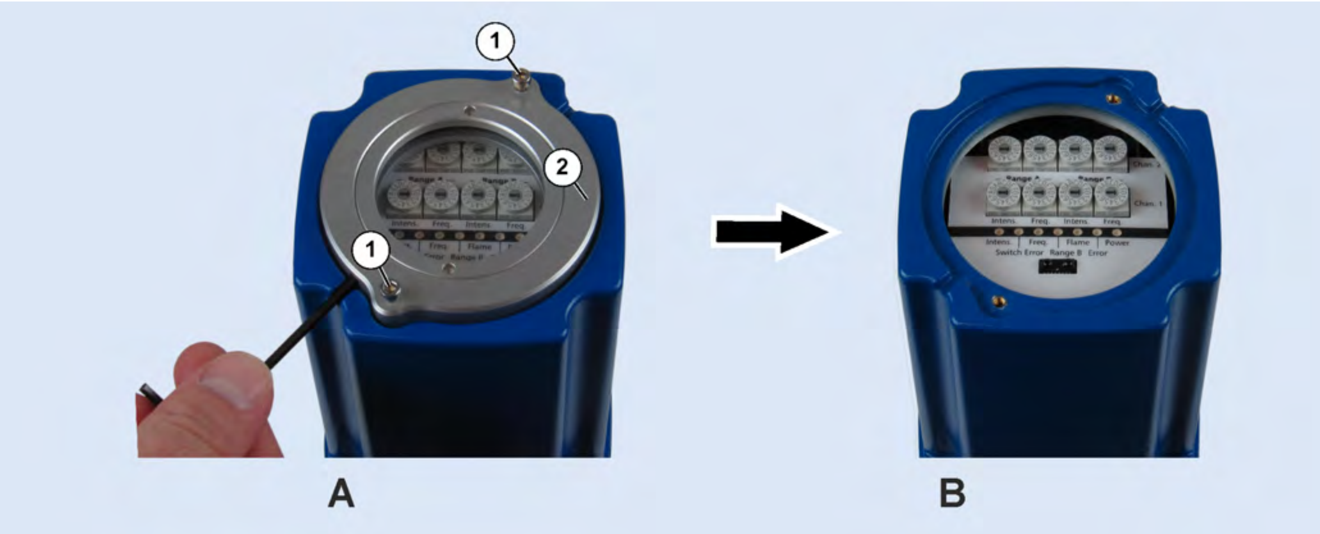


Fig. 13.1: Remove the plug-in cover (figure shows M5 housing with metal plug-in cover). A: Flame monitor with plug-in cover, B: Flame monitor without plug-in cover

1. Undo the two socket head screws (1).
2. Remove the plug-in cover (2).

If the plug-in cover sticks, the recesses can be used to lift off the plug-in cover using a socket head wrench.

13.2 Screwing down the plug-in cover (housing M5)

M5

Fig. 13.2: Push the plug-in cover into the ring groove (figure shows M5 housing with metal plug-in cover).

1. Carefully place the plug-in cover onto the cover. Press it into the circular opening in the cover, making sure that the plug-in cover does not tilt.

2. Screw down the plug-in cover with the two socket head screws. Otherwise, the flame monitor's specified IP protection will not be achieved.

13.3 Removing the cover (housing M5)

M5

DANGER

Danger of death due to electric current!

An immediate danger of death exists in case of contact with live parts. Electronic components may be damaged by a short-circuit. Therefore:

- Disconnect the flame monitor before removing the cover. When doing so, take into account both the flame monitor's supply voltage and the supply voltage of the contact circuit.
- Only ever have work on the opened device performed by specialised electricians.

NOTICE

Damage to electronic components due to electrostatic discharge (ESD)

Electronic components are becoming smaller and smaller and more and more complex. Their susceptibility to damage from electrostatic discharge is increased accordingly. Therefore:

- To protect the components, measures must be undertaken to prevent electrostatic discharge during all work performed at the open device (ESD protection).
- To prevent static charges building up on the human body, service employees can for example be equipped with a personal earthing system.

Ex M5

In hazardous areas:

WARNING

Risk of explosion due to open sources of ignition!

The housing contains sources of ignition. Opening the housing in a potentially explosive atmosphere can lead to explosions. Therefore, before opening:

- Make sure that there is no potentially explosive atmosphere.
- If there is a potentially explosive atmosphere, disconnect the flame monitor. When doing so, take into account both the flame monitor's supply voltage and the supply voltage of the contact circuit.

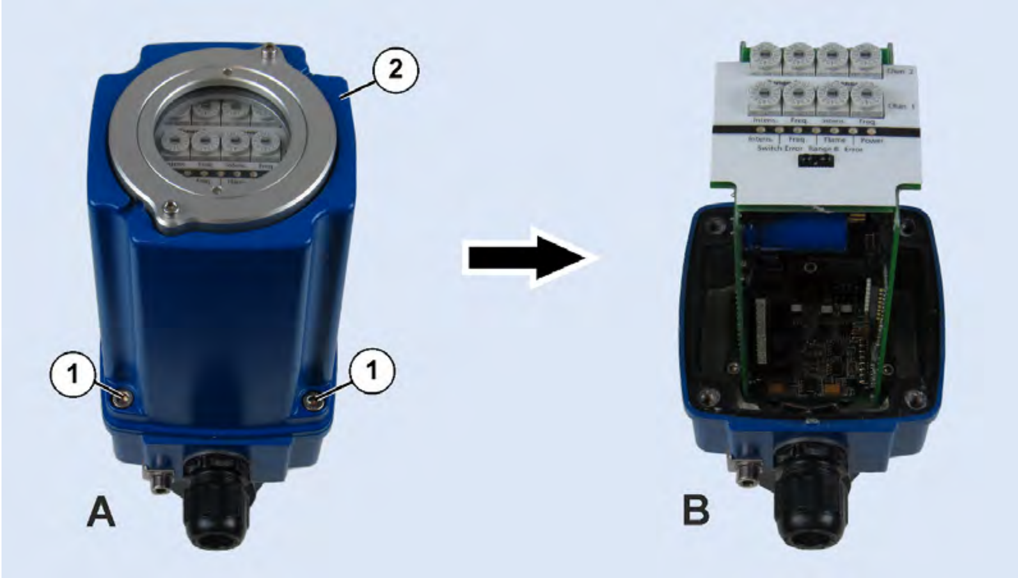


Fig. 13.3: Remove the cover (figure shows housing M5). A: Flame monitor with cover, B: Flame monitor without cover

1. Undo the four socket head screws (**1**).
2. Remove the cover (**2**). To do so, carefully lift it straight upwards without undoing or damaging the printed circuit boards within.

13.4 Screwing down the cover (housing M5)

M5

Fig. 13.4: Screw down the cover (figure shows housing M5).

1. Make sure that the cover is correctly aligned (**1**).
2. Slide the cover (**2**) carefully over the printed circuit boards.
3. Screw in the four socket head screws (**3**).

13.5 Opening housing M4 (for zone 1/21)

DANGER

Danger of death due to electric current!

An immediate danger of death exists in case of contact with live parts. Electronic components may be damaged by a short-circuit. Therefore:

- Disconnect the flame monitor before removing the cover. When doing so, take into account both the flame monitor's supply voltage and the supply voltage of the contact circuit.
- Only ever have work on the opened device performed by specialised electricians.

WARNING

Risk of explosion due to open sources of ignition!

The housing contains sources of ignition. Opening the housing in a potentially explosive atmosphere can lead to explosions. Therefore, before opening:

- Make sure that there is no potentially explosive atmosphere.
- If there is a potentially explosive atmosphere, disconnect the flame monitor. When doing so, take into account both the flame monitor's supply voltage and the supply voltage of the contact circuit.

M4

Fig. 13.5: Safety screw

1. Undo the safety screw (socket head) by turning a few rotations in an anti-clockwise direction. The screw is permanently connected, i.e. it cannot be completely unscrewed by being turned in this direction.

Fig. 13.6: Undoing the cover

2. Undo the cover (GH) by turning it in an anti-clockwise direction. Do not use any tools (e.g. wrenches, water pump pliers etc.). Even if the housing shape appears suitable for the use of these kind of tools, they could damage the fine thread or the sealing face. Even small score marks in these gaps and sealing faces are enough to mean that the explosion protection will no longer be guaranteed. In this case, the housing parts must always be replaced.

Fig. 13.7: Removing the cover

3. Carefully remove the cover backwards in a straight line. The cover must only be swivelled off the axis once the printed circuit boards are completely exposed (see Fig. 13.9).

Fig. 13.8: Do not tilt the cover when removing

4. Hold the cover securely when removing; it is unexpectedly heavy! Do not allow the cover to slide along the printed circuit boards; **the printed circuit boards are not suitable support aids for the cover!**

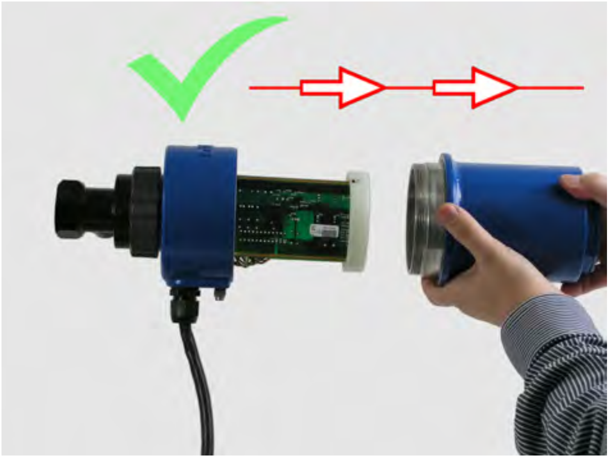


Fig. 13.9: Do not allow the cover to slide along the printed circuit boards

13.6 Closing housing M4 (for zone 1/21)

M4

Fig. 13.10: Do not tilt the cover when sliding it on

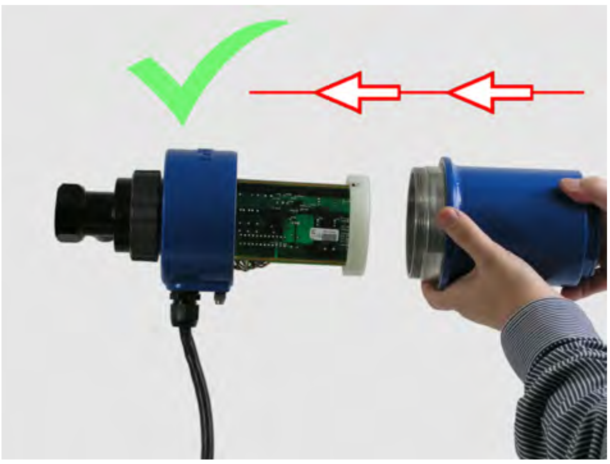


Fig. 13.11: Move the cover onto the longitudinal device axis when sliding it on

1. Slide on the cover in a straight line from behind. Move the cover onto the longitudinal device axis while doing so. Do not allow the cover to rest on the printed circuit boards; **the printed circuit boards are not suitable support aids for the cover!**

Fig. 13.12: Move the cover onto the longitudinal device axis when sliding it on

2. Carefully screw on the cover (**GH**) by turning it in a clockwise direction. The screw connection is established via a thread, which also creates the pressure-tight seal. It has proved useful to start by screwing on the cover with light pressure in an anti-clockwise direction until the thread guide noticeably subsides. Then screw on in a clockwise direction until the cover is lying securely on the base. This is the case when the safety screw is right next to the red dot on the Ex middle part.

Fig. 13.13: Correctly closed housing

NOTICE

If the thread is positioned correctly in the guide, the cover can be easily screwed on. **No tools are required** for this purpose!

Never use force to screw in the thread. This can damage the thread.

3. Screw down the safety screw (socket head) by turning a few rotations in a clock-wise direction, thereby securing the cover against being accidentally undone.

14 Setup

The operator is responsible for setting up the flame monitor for the specific plant in a way that ensures that no incorrect or dangerous plant states can occur. This relates to both the positioning of the flame monitor and its alignment to the flame, as well as the flame monitor settings.

NOTICE

Damage to electronic components due to electrostatic discharge (ESD)

Electronic components are becoming smaller and smaller and more and more complex. Their susceptibility to damage from electrostatic discharge is increased accordingly. Therefore:

- To protect the components, measures must be undertaken to prevent electrostatic discharge during all work performed at the open device (ESD protection).
- To prevent static charges building up on the human body, service employees can for example be equipped with a personal earthing system.

The plug-in cover will need to be removed in order to adjust the flame monitor settings. Information on this can be found in section 13.1 Removing the plug-in cover (housing M5) [▸ 55].

The cover will need to be removed in order to adjust the flame monitor settings. Information on this can be found in section 13.5 Opening housing M4 (for zone 1/21) [▸ 58].

14.1 Set switching thresholds

Defining the **switching threshold** specifies whether the size of the present flame signal [▸ 101] should lead to a *flame ON* or *flame OFF* signal. Only flame signals with higher pulse rates than that of the switching threshold are evaluated as *flame ON*.

The switching threshold can be adjusted during operation using rotary switches. The switch position **0** corresponds to the highest response threshold, and the flame sensor will need to generate the largest flame signal (the highest pulse rate) for a flame ON notification. The switch position **9** is the lowest response threshold, and a small flame signal can hold or actuate a flame ON notification from the flame monitor.

WARNING

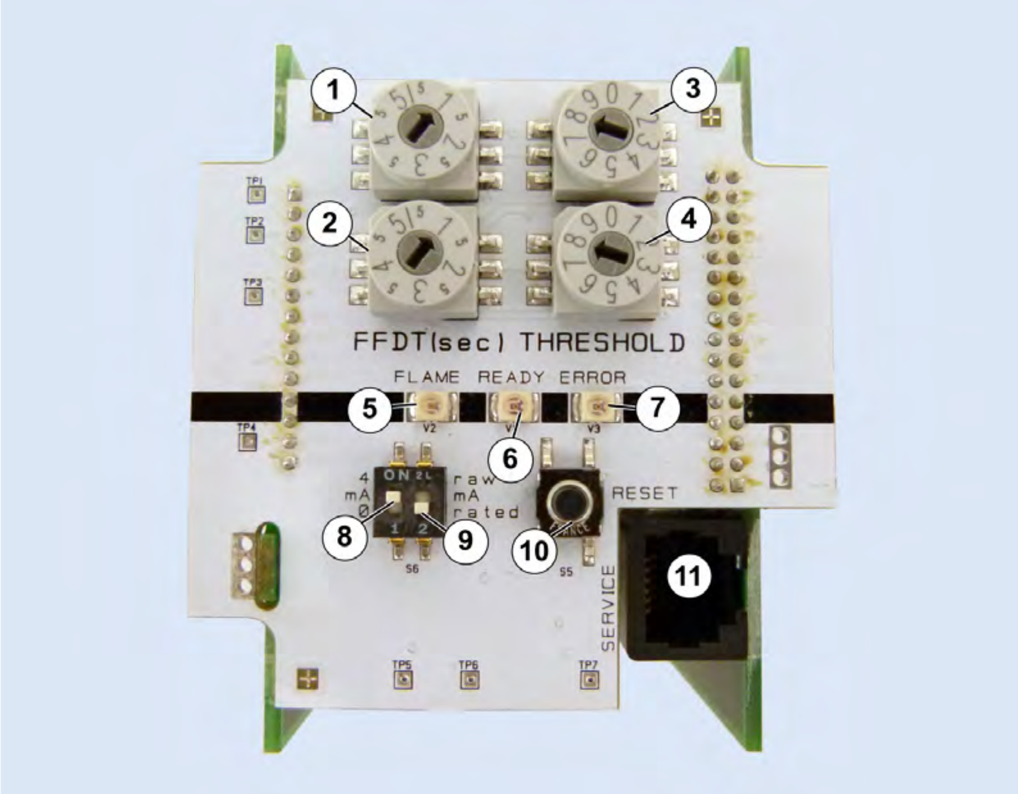
Wrong evaluation of the flame signal due to incorrectly set switching threshold!

If the switching threshold is incorrectly set, the flame monitor will not show the burner state correctly.

Therefore:

- Secure the flame monitor against unauthorised modification of the switching threshold!

The rotary switches do not have end stops. You can continue to rotate them to move from switch position **9** to switch position **0**.



1, 2	Rotary switches; FFDT
3, 4	Rotary switches; threshold
5	LED **FLAME** (green): The flame monitor reports flame ON
6	LED **READY** (yellow): **Permanently illuminated:** The flame monitor is ready for operation **Flashing:** The flame monitor has a fault. Information on the type of fault and rectification measures can be found in Section 17 Fault rectification [▸ 75].
7	LED **ERROR** (red): **Permanently illuminated:** An error has occurred **Flashing:** The rotary switches have different settings in at least one of the two pairs of rotary switches. This setting difference is causing an error shutdown after 8 s.
8	Switch 0 … 20 mA /4 … 20 mA *(0/4)*
9	Switch for evaluated/raw flame signal *(rated/raw)*
10	Reset
11	Service interface

Fig. 14.1: Front panel

1. Set the rotary switches **3** and **4** to a suitable identical value within 8 seconds.

✓ If the rotary switch values do not match, the LED **Error** (**7**) will flash. After 8 seconds, the flame monitor will perform an error shutdown.

> You will find detailed information about finding out an appropriate switching threshold in the following chapter.

After setting the switching threshold:

1. Securely screw on the plug-in cover (see Section 13.2 Screwing down the plug-in cover (housing M5) [▸ 55]).

✓ This is the only way of achieving the flame monitor's specified IP protection class. The flame monitor must not be operated when open!

1. Close the housing (see section 13.6 Closing housing M4 (for zone 1/21) [▸ 60]).

✓ Otherwise, the flame monitor will not provide the required explosion protection.

14.2 Determine suitable switching threshold

The flame monitor must be carefully adapted to the particular burner properties in order to ensure successful operation. The objectives of the flame monitor setting adjustment are set out below. The description of the procedure for adjusting the settings is then provided.

14.2.1 Objectives

With optimal settings, the flame monitor must reliably report a switched-off burner or an impermissible impairment of a flame pattern. The influence of neighbouring flames or other external light sources must not generate a defective plant status.

In practice, static emitters such as glowing boiler components are modulated by combustion air currents or flue gas vapours, and act as background radiation with dynamic radiation parts. If this radiation is within the capture range, i.e. in the IR range and the usual flicker frequency range (approx. 10 to 200 Hz) for a flame, an output signal is generated by the flame monitor according to the intensity and the dynamic of the received radiation. The flame monitor's switching threshold must not be exceeded as a result of this background radiation, and must never lead to a flame ON notification.

Fig. 14.2: LEDs on the front panel: LED FLAME (1) is off.

Flame OFF

When the burner is switched off and with maximum background radiation, the switching threshold is set high enough to ensure that the flame monitor will reliably issue a flame OFF notification. This means:

- The LED FLAME (1) is off.
- The value of the analogue output for the flame intensity is lower than 8 mA (4 … 20 mA) or 5 mA (0 … 20 mA)
- The relay output for the flame OFF signal is closed.
- The relay output for the flame ON signal is open.

Flame ON

When the burner is switched on, the flame monitor should reliably exceed the set switching threshold and issue a flame ON notification. This means:

- The LED FLAME (1) lights up green.

- The value of the analogue output for the flame intensity is higher than 12 mA (4 … 20 mA) or 10 mA (0 … 20 mA)
- The relay output for the flame ON signal is closed.
- The relay output for the flame OFF signal is open.

14.2.2 Procedure

The flame sensor, which is part of the flame monitor, converts the light signal from the flame into a pulsed electrical signal. The pulse rate is indicative of the flame intensity. This raw flame signal (flame sensor pulse rate) is required in order to adjust the flame monitor settings.

The display unit D-ZS 087-20 (accessory) provides assistance when adjusting the settings. It displays the measured minimum and maximum pulse rate of the flame sensor over a period of time. The resulting suitable switch position for setting the switching threshold is also displayed.

An **amperemeter** can be used to display the raw flame signal via the analogue current output (see also Section 19 Circuit diagrams [▸ 78]) if you do not have a display device.

To work out the switching threshold, always use the statuses of **flame OFF with maximum background radiation** (e.g. the burner being monitored is switched off; all other burners are switched on) and **flame ON with minimum background radiation** (e.g. the burner being monitored is switched on; all other burners are switched off)

1. Use the switch (**8** in Fig. 14.1 in Section 14.1 Set switching thresholds [▸ 62]*)* to select position *0* for the analogue output 0 … 20 mA.
2. Use the switch (**9**) to select the position *raw* for the raw flame signal.
   - ✓ The pulse rates from 0 to 2047 pulses/s are displayed in linear format at 0 … 20 mA. If the analogue output displays 15.00 mA, for example, this corresponds to a flame sensor pulse rate of around 1500 pulses/s.
3. Generate the state **flame OFF with maximum background radiation.** To do so, switch off the burner being monitored and switch on all other burners.
4. Observe the pulse rate of the raw flame signal for around one minute. Over this period, work out the highest and lowest flame signal pulse rate with the burner switched off.
   - ✓ A possible result for flame OFF could be (example!): 900 … **1300** pulses/s
5. Generate the state **flame ON with minimum background radiation.** To do so, switch on the burner being monitored and switch off all other burners.
6. Observe the raw flame signal for around one minute. Over this period, work out the highest and lowest flame signal pulse rate with the burner switched on.

✓ A possible result for flame ON could be (example!): **1800** … 2500 pulses/s

Working out the switching threshold

To select the switching threshold, it is important to know the maximum pulse rate with the burner switched off and the minimum pulse rate with the burner switched on. The switching threshold should be in between these two values.

Possible switching thresholds are shown in the following table. As the switching thresholds are subject to hysteresis, one switch-on threshold and one switch-off threshold is specified.

Rotary switch	Start of range 0/4 mA	Switch-off threshold 5/8 mA	Switch-on threshold 5/8 mA	End of range 20 mA
0	2048	2556	2810	4080
1	1536	1984	2208	3328
2	1024	1432	1636	2656
3	768	1088	1248	2048
4	608	892	1034	1744
5	384	624	744	1344
6	256	448	544	1024
7	128	288	368	768
8	64	132	166	336
9	32	56	68	128

Table 14.1: Switch positions with the associated switching thresholds

The numerical values shown are the pulse rates of the flame signal. Signals with pulse rates below the start of the range are shown as 0/4 mA signals, and signals with pulse rates above the end of the range are shown as 20 mA signals.

If the flame monitor displays flame OFF and the pulse rate of the flame signal is increasing, the flame monitor will switch to flame ON when the switch-on threshold pulse rate is reached. If the pulse rate of the flame signal is decreasing when in the flame ON state, the flame monitor will switch to flame OFF when the switch-off threshold pulse rate is reached.

For the above example, the switch-on threshold must be below 1800 Hz and the switch-off threshold above 1300 Hz in order that the flame monitor can correctly detect the burner state. The associated rotary switch settings for suitable switching thresholds can be found in the table. Only switch position 2 is suitable for the above example.

1. Read the suitable switch positions for your measured pulse rates from the table.
   ✓ Multiple switch positions may be possible.

WARNING

Wrong evaluation of the flame signal due to incorrectly set switching threshold!

If the switching threshold has been set too low, the flame monitor will show flame ON even though the burner is switched off.

Therefore:

- Make sure that flame OFF is reported in all operating states and in an extremely wide range of firing conditions when the burner is switched off.
- Prevent faults as a result of background radiation.

2. In general, select a medium switch position so that the difference between the switch-on threshold and the minimum pulse rate when the burner is switched on is roughly the same as the difference between the switch-off threshold and the maximum pulse rate when the burner is switched off.

> If the flame signal temporarily fluctuates more sharply when the burner is switched on and the flame monitor reports flame OFF, it may help to slightly reduce the switching thresholds. To do so, use the rotary switch to select a slightly higher value.

3. Set the required switch position on the device (see Section 14.1 Set switching thresholds [▶ 62]).
4. Adjust the analogue output if necessary (see Section 14.4 Setting the analogue output [▶ 67]).

14.3 Set flame failure detection time (FFDT)

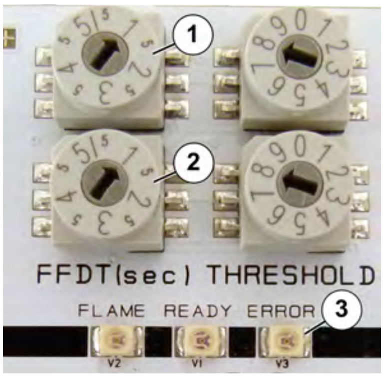
The **Flame Failure Detection Time (FFDT)** (formerly: safety time) is the time it takes the flame monitor to respond to a flame signal failure, e.g. as a result of an impaired combustion process (see also Section 9 Functional description [▶ 39]).

The FFDT is factory-set to 1.0 s as standard. The FFDT can be set to a value of between 0.5 s and 5.0 s in 0.5-s intervals using the rotary switches.

However, some approvals pose a limit on the selection of the FFDT:

Devices with

- **FM approval** (regional code: H): Select max. 4 s for FFDT!
- **AGA approval** (regional code: C) and for simultaneous use in Australia: Select max. 2 s for FFDT!



1. Set the required FFDT value using the rotary switches (**1**, **2**).

✓ If the rotary switch values do not match, the LED Error (**3**) will flash. After 8 seconds, the flame monitor will perform an error shutdown.

M5

1. Securely screw on the plug-in cover (see Section 13.2 Screwing down the plug-in cover (housing M5) [▶ 55]).

✓ This is the only way of achieving the flame monitor's specified IP protection class. The flame monitor must not be operated when open!

M4

1. Close the housing (see section 13.6 Closing housing M4 (for zone 1/21) [▶ 60]).

✓ Otherwise, the flame monitor will not provide the required explosion protection.

14.4 Setting the analogue output

The raw flame signal (pulse frequency of the flame sensor) and the evaluated flame signal are optionally available as an analogue output (4 … 20 mA or 0 … 20 mA).

The **evaluated flame signal** is dependent on the pulse rate of the flame sensor's flame signal and the set switching threshold [▸ 101]. When the switching threshold is adjusted, this automatically defines the range for the pulse rate (see Table 14.1 on page [▸ 65]). This range is displayed at the analogue output in linear format.

During operation, recording the evaluated flame signal over longer periods of time can provide useful information on the state of the burner.

To configure the analogue output, proceed as follows:

1. Use the switch (**8** in Fig. 14.1 in Section 14.1 Set switching thresholds [▸ 62]) to select 0 … 20 mA or 4 … 20 mA for the analogue output range.
2. Use the switch (**9**) to select the position *rated* for the evaluated flame signal.

✓ The evaluated flame signal is provided as an analogue output (for the contacts, see the circuit diagram in Section 19 Circuit diagrams [▸ 78]).

M5

1. Securely screw on the plug-in cover (see Section 13.2 Screwing down the plug-in cover (housing M5) [▸ 55]).

✓ This is the only way of achieving the flame monitor's specified IP protection class. The flame monitor must not be operated when open!

M4

1. Close the housing (see section 13.6 Closing housing M4 (for zone 1/21) [▸ 60]).

✓ Otherwise, the flame monitor will not provide the required explosion protection.

The pulse rates for the **raw flame signal** are displayed at the analogue output in linear format. The raw flame signal is required when adjusting the flame monitor settings (see Section 14.2.2 Procedure [▸ 65]).

15 Operation

Changes to the firing plant, the use of different fuels or new processes (e.g. light load) may mean that the flame monitor settings are no longer correct:

WARNING

Wrong evaluation of the flame signal due to incorrectly set flame monitor!

An incorrectly set flame monitor will provide incorrect indications of the burner states.

Therefore:

- Make sure that the flame monitor reliably detects the burner states of flame ON/OFF **in all operating states.** Information on modifying the flame monitor to meet plant-specific conditions can be found in Section 14 Setup [▶ 62].
- Secure the flame monitor against unauthorised modifications being made to its alignment to the flame and to the selected settings on the device.

15.1 Switching the system off

1. Switch off the operating voltage for the flame monitor.

NOTICE

Risk of lens contamination and risk of the flame monitor overheating due to failure of the purge air

The purge air serves to protect the system fitted onto the burner. The purge air shields the components from hot and/or aggressive gases and dust. A flame monitor with a contaminated lens will no longer be able to reliably detect flames. Therefore:

- Make sure to continually supply the flame monitor with purge air even when switched off.
- Make sure that a purge air failure is detected immediately.

✓ You can switch the flame monitor back on again at any time without firstly having to clean the lens.

15.2 Switching on

1. Make sure that the flame monitor lens is clean. A flame monitor with a contaminated lens will not be able to reliably detect flames.

The lens and the burner sighting tube may be contaminated if the flame monitor was not supplied with purge air when switched off, e.g. during a revision. Information on cleaning the lens can be found in Section 16.2 Carrying out a visual inspection [▶ 70]. The flame monitor can then be put back into operation (see Section 12 Initial commissioning [▶ 54]).

2. Make sure that the flame monitor was continually supplied with purge air when switched off.
3. Switch on the operating voltage.

✓ Once initialisation is complete, the flame monitor is ready for operation. The settings that were in place prior to switch-off remain unchanged.

16 Maintenance

16.1 Maintenance schedule

The operating company defines the device-specific maintenance schedule . This takes account of the local climate and ambient conditions, contamination at the location as well as the operating company's individual maintenance regulations.

The following table can only be used as a basis for the device-specific maintenance schedule under normal operating conditions:

Interval	Maintenance work	Performed by
6 months, adapted to the plant conditions	Visual inspection (see Section 16.2 Carrying out a visual inspection [▸ 70])	Specialised electrician
After 10,000 operating hours:	Replace the UV cell (see Section 16.3 Replacing the UV cell (D-LX 110 UL***) [▸ 71])	

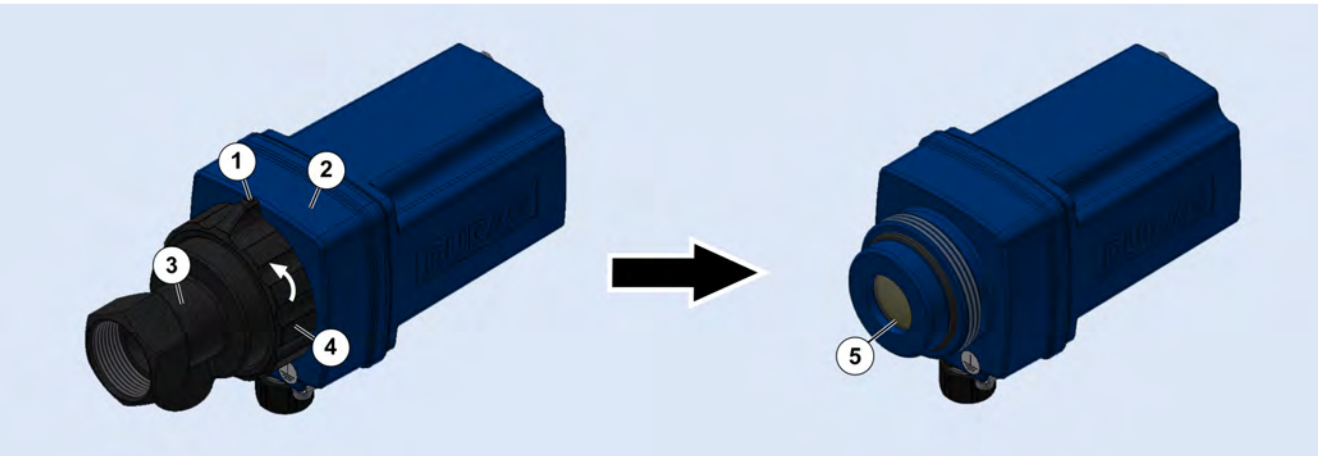
Table 16.1: Maintenance schedule

If the flame monitor is not sufficiently maintained, or is not maintained at all, a burner failure should be expected.

16.2 Carrying out a visual inspection

Based on the local conditions and the burner type, make the necessary preparations to allow for the unscrewing of the flame monitor. In particular, all adjacent components must be cooled to room temperature. The supply voltage for the flame monitor can remain switched on.

1. Check the purge air supply and the hose for damage.
2. Check the hose connections for leak-tightness and secure fit.
3. Check the flame monitor housing. The housing must be leak-tight and intact. The plug-in cover must be fitted.



4. Check the lens (**5**) of the flame monitor for contamination. To do so, detach the front part (**3**) from the middle part (**2**). To do so, undo the safety screw (**1**). Unscrew the connecting nut (**4**).

5. If necessary, remove any visual deposits from the lens (**5**) using oil-free compressed air or a soft lint-free cloth (lens cloth or microfibre cloth). Use water or alcohol as well if necessary.

Adjust the purge air volume if there is regularly contamination on the outside of the lens.

6. Screw the front part (**3**) onto the middle part (**2**) by hand with the connecting nut (**4**).
7. In order to prevent the middle part becoming loose unintentionally, the connecting nut is fitted with a locking screw (**1**). Tighten the locking screw.

✓ The flame monitor is ready for operation again.

16.3 Replacing the UV cell (D-LX 110 UL***)

Qualified personnel: Specialised electrician

Required: **D-LX 110 UL SP UV photocell** (article no.: 4 017 786)

D-ZS 077-10 Testing light source for the UV-C range for function checking (article no.: 1 107 760)

The UV cell ages even if not being used. Therefore:

- Store the UV cell for a maximum of four years.

NOTICE

Damage to electronic components due to electrostatic discharge (ESD)

Electronic components are becoming smaller and smaller and more and more complex. Their susceptibility to damage from electrostatic discharge is increased accordingly. Therefore:

- To protect the components, measures must be undertaken to prevent electrostatic discharge during all work performed at the open device (ESD protection).
- To prevent static charges building up on the human body, service employees can for example be equipped with a personal earthing system.

1. Disassemble the flame monitor (see Section 18.1 Disassembly [▶ 77]). Move the flame monitor to a clean location and a ESD-suitable workstation.
2. Open the housing (see Section 13 Opening and closing the housing [▶ 55]).

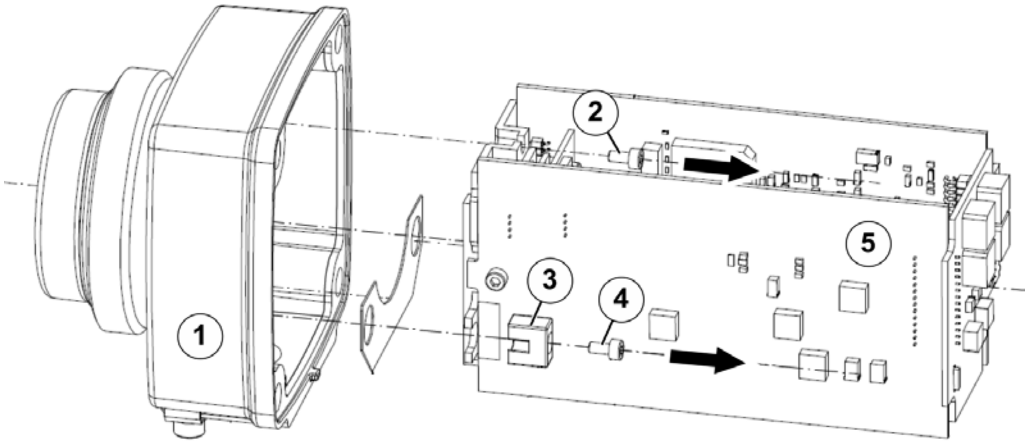


Fig. 16.1: Disassemble the flame monitor (the model shown is the D-LX 110 in housing M5)

3. Detach the insert (5) from the middle part (1). To do so, undo the screws (2) and (4). Remove the plastic clip (3). Proceed in exactly the same way on the opposite side.

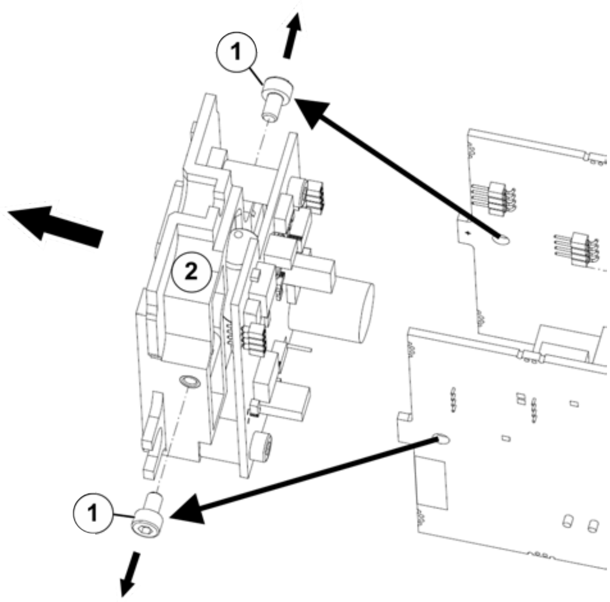


Fig. 16.2: Disassemble the old UV cell

4. Undo the screws (1). Remove the UV cell (2) from the insert.

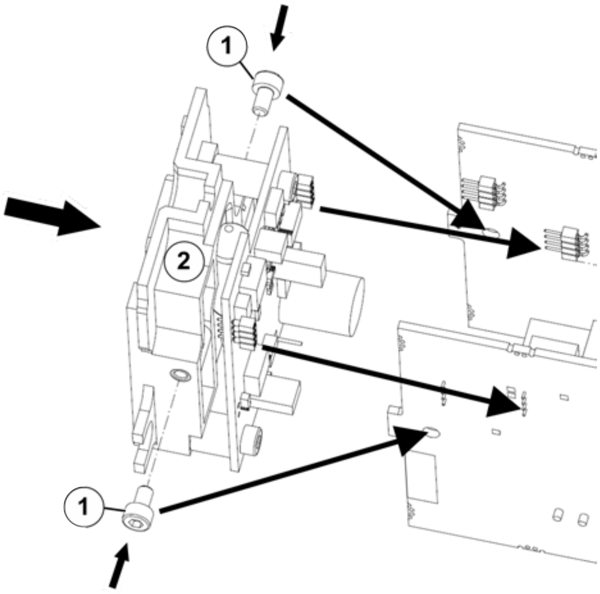


Fig. 16.3: Fitting a new UV cell

5. Carefully push the new UV cell (**2**) onto the plug-in contacts. Tightly screw the UV cell onto the insert using the screws (**1**).

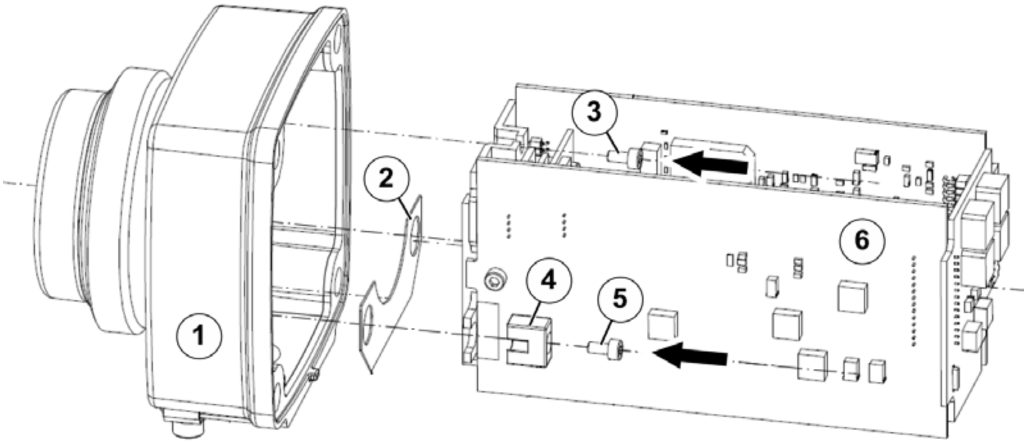


Fig. 16.4: Assemble the flame monitor (the model shown is the D-LX 110 in housing M5)

6. Attach the insert (**6**) to the middle part (**1**): Make sure that the insulation plate (**2**) is positioned in the middle part (**1**) with the two holes above the mounting domes. Place the insert (**6**) onto the mounting domes. Attach the plastic clip (**4**) over the mounting dome. Tightly screw down the insert using the screws (**3**) and (**5**). Proceed with the plastic clip in exactly the same way on the opposite side.
7. Close the housing (see section 13 Opening and closing the housing [▸ 55]).

Checking the function of the flame monitor using the testing light source (D-ZS 077-10)

1. Screw the flame monitor (**1**) onto the testing light source (**2**).
2. Connect the flame monitor to the power supply (see Section 11.4 Establishing electrical connections [▸ 47]).
3. Set the switching threshold for the intensity to **9** (see Section 14.1 Set switching thresholds [▸ 62]). Note down the previously used setting if necessary.

4. Make sure that the testing light source is switched off.
   ✓ The flame monitor must report **flame OFF**: The LED FLAME (**1**) is **off**.
5. Switch on the testing light source using the switch (**1**).
   ✓ The flame monitor must report **flame ON**: The LED FLAME (**1**) is **on**.
6. Switch off the testing light source.
   ✓ The flame monitor must report **flame OFF**: The LED FLAME (**1**) is **off**.

✓ The flame monitor has always correctly detected the **flame ON/OFF** signal: The flame monitor has passed the function check. Set the intensity switching threshold used in the past. The flame monitor can be re-installed (see Section 11 Installation [▸ 44]).

The flame monitor has incorrectly detected the **flame ON/OFF** signal at least once: The flame monitor has not passed the function check. Do not use the flame monitor! DURAG Service will gladly provide you with further assistance.

17 Fault rectification

Fault	Possible reason	How to rectify the fault
M4 M5 General abnormal behaviour, e.g. faulty flame monitor	Interfering influences due to compensating currents on the earth conductor, e.g. currents from an ignition device against earth potential	Install an isolator D-ZS 117-I between the sighting tube and the flame monitor (see also Section 11.1 Useful accessories [▸ 44]).
The flame monitor is driven into saturation: Flame ON is reported for all switching threshold settings, even when the burner is switched off.	The flame signal is very intensive.	Contact DURAG Service and insert an aperture (see Section 8.6 Accessories [▸ 34]) into the purge air flange if necessary.
The analogue output is imprecise, e.g. doesn't reach the maximum value.	The load is too high The supply voltage is too low	Select the load and supply voltage as explained in Section 7 Technical data [▸ 27]
		Contact DURAG Service.
No message, the flame monitor is switched off.	Fuse F1 has blown; the supply voltage was incorrectly polarised, for example	Rectify the fault that caused the fuse to blow, replace fuse F1 (see Section 17.1 Replacing the electrical fuse F1 [▸ 76])
No message at the relay output. The flame monitor is switched on.	Fuse F3 has blown.	Rectify the fault that caused the fuse to blow, contact DURAG Service

Table 17.1: Fault table

The flame monitor has a fault

The red LED **ERROR** lights up (see Section 8.5 Front panel (all types) [▸ 32]) and the relays for the flame ON notification and the ready for operation notification drop out. The type of fault is indicated by the LED **READY**:

LED READY (yellow) flashes in quick succession	How to rectify the fault
Once	Contact DURAG Service.
Twice	
3 times	**Cause:** The rotary switches have different settings in at least one of the two pairs of rotary switches. **Action:** Set the rotary switches within a pair to the same value. Press the Reset button (see Section 8.5 Front panel (all types) [▸ 32]).
4 times	Press the **Reset** button (see Section 8.5 Front panel (all types) [▸ 32]) and switch off the supply voltage if necessary.
5 to 9 times	Press the **Reset** button (see Section 8.5 Front panel (all types) [▸ 32]).

Table 17.2: Fault messages

The actions should usually allow the flame monitor to continue functioning once it has been restarted. Otherwise, DURAG Service will gladly provide you with further assistance.

NOTICE

The flame monitors (including the Ex versions and all associated components) are devices with type approvals.

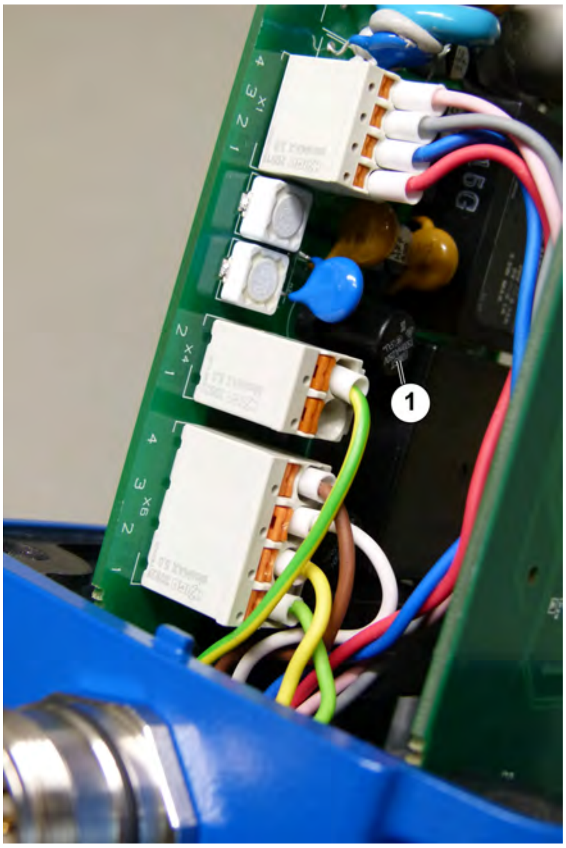
Any changes to to devices with type approvals will lead to the loss of the approval. Repairs must only be performed by the manufacturer or persons that it has authorised to do so. This is the only way of ensuring responsible compliance with the specifications for the type approval of the certifying body.

17.1 Replacing the electrical fuse F1

Qualified personnel: Specialised electrician

Required: Electrical fuse, type: 0.5 A delayed action MST 250

1. Remove the cover (see Section 13 Opening and closing the housing [▶ 55]).



2. Replace the electrical fuse F1 (1).
3. Close the housing (see section 13 Opening and closing the housing [▶ 55]).

18 Dismantling and disposal

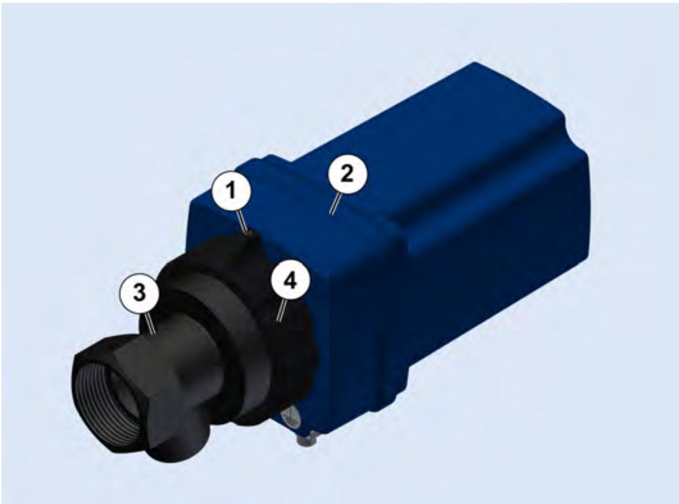
18.1 Disassembly

Based on the local conditions and the burner type, make the necessary preparations to allow for the unscrewing of the flame monitor. In particular, all adjacent components must be cooled. The flame monitor must be switched off (see Section 15.1 Switching the system off [▸ 69]).

Electrical work must be performed by a specialised electrician (see Section 2.4 Qualified personnel [▸ 11]).

Electrical work in hazardous areas must be performed by a specialised electrician with additional knowledge of explosion protection (see Section 2.4 Qualified personnel [▸ 11]).

1. Switch off the purge air supply. Disconnect the flame monitor from the purge air line.
2. Disconnect the connecting cable from the plant.



3. Undo the safety screw (1). Unscrew the connecting nut (4).
   ✓ The middle part (2) can be detached from the front part (3) together with the cover.
4. Unscrew the front part (3) from the sighting tube.

18.2 Disposal

Dispose the product in an environmentally friendly manner. Observe the national valid regulations.

19 Circuit diagrams

19.1 Devices with connecting cable and cable gland (models D-LX ***/*CG)

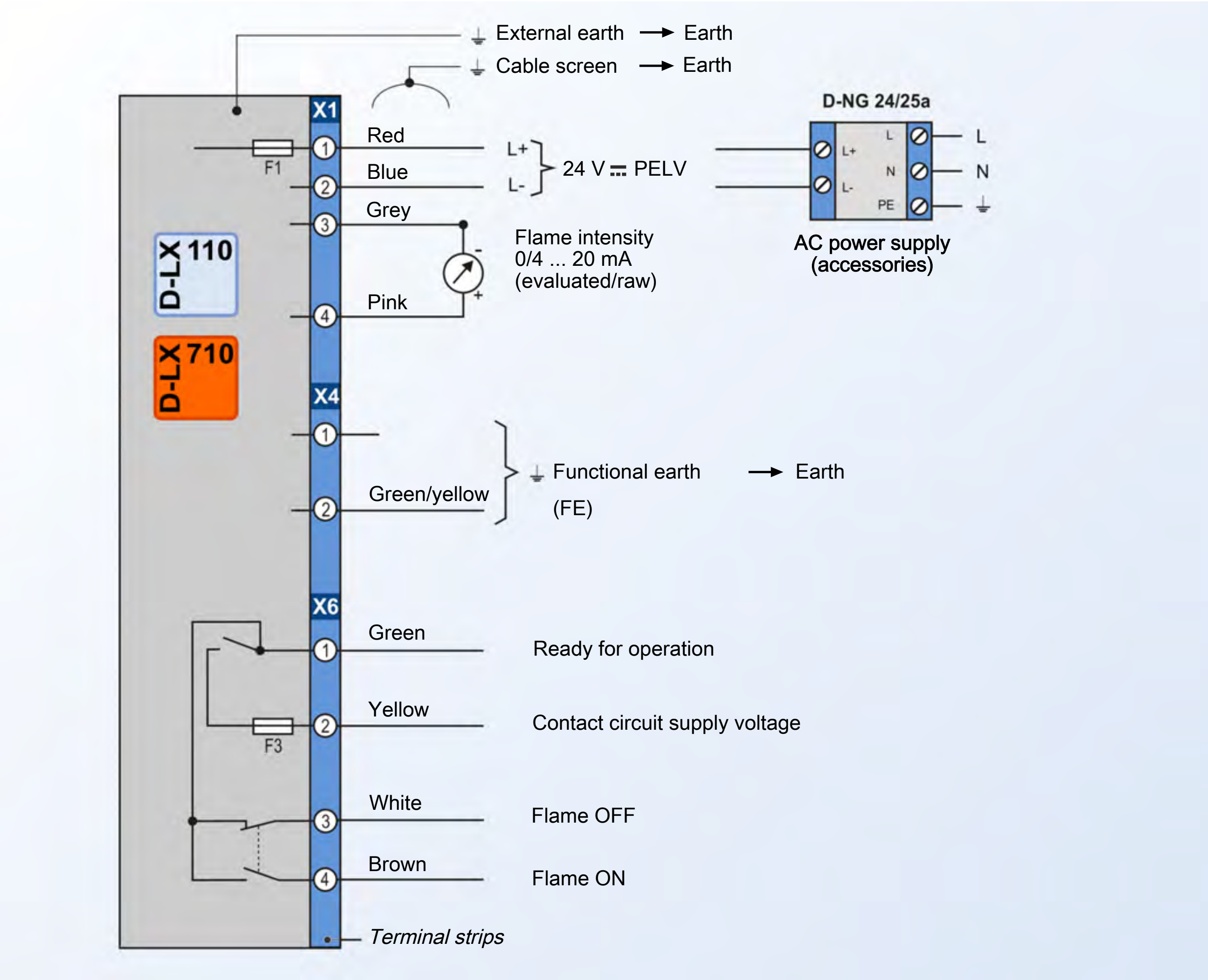


Fig. 19.1: Circuit diagram for D-LX 110/710 with connecting cable

19.2 Devices with panel plug (models D-LX ***/MP7)

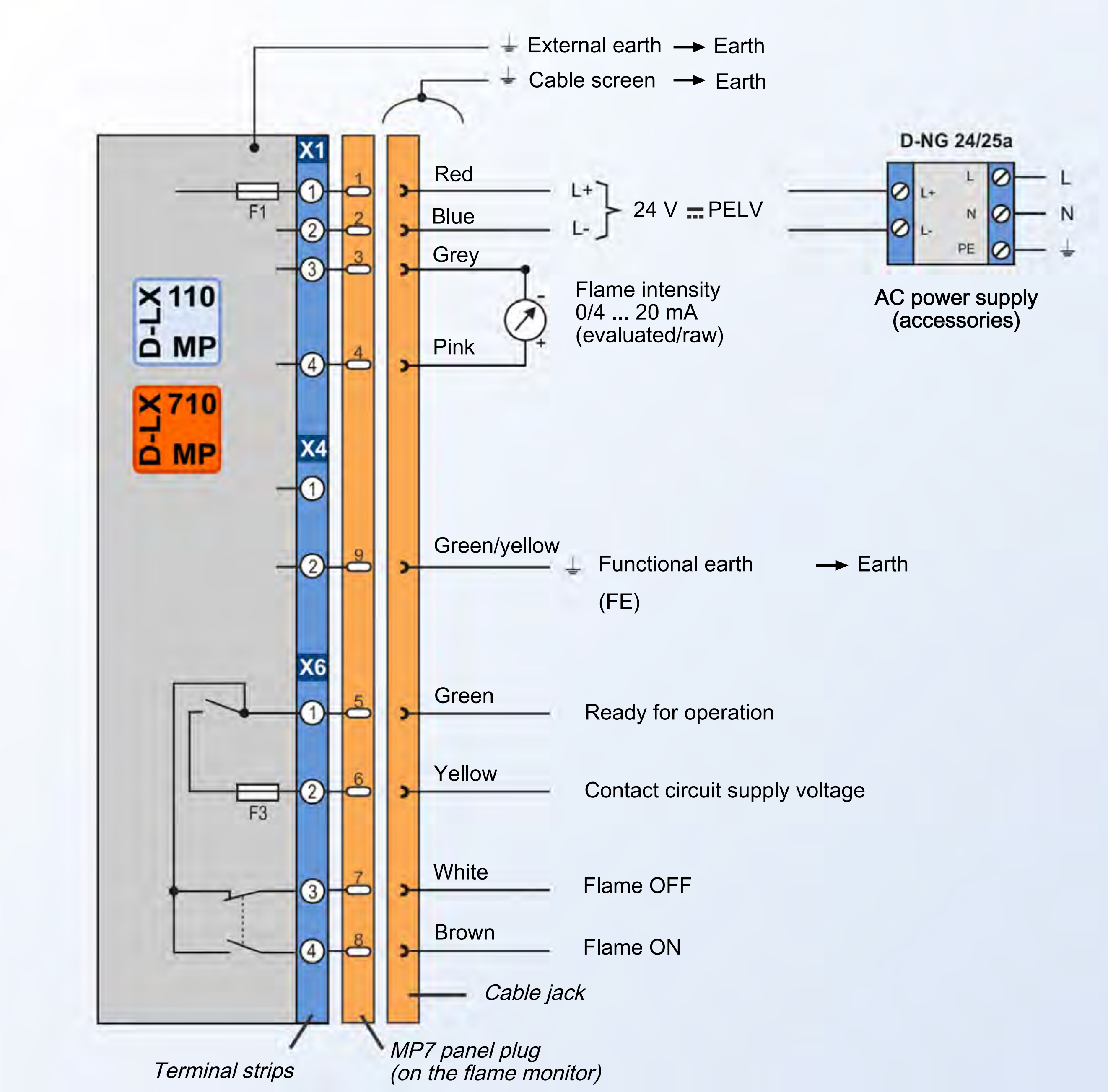
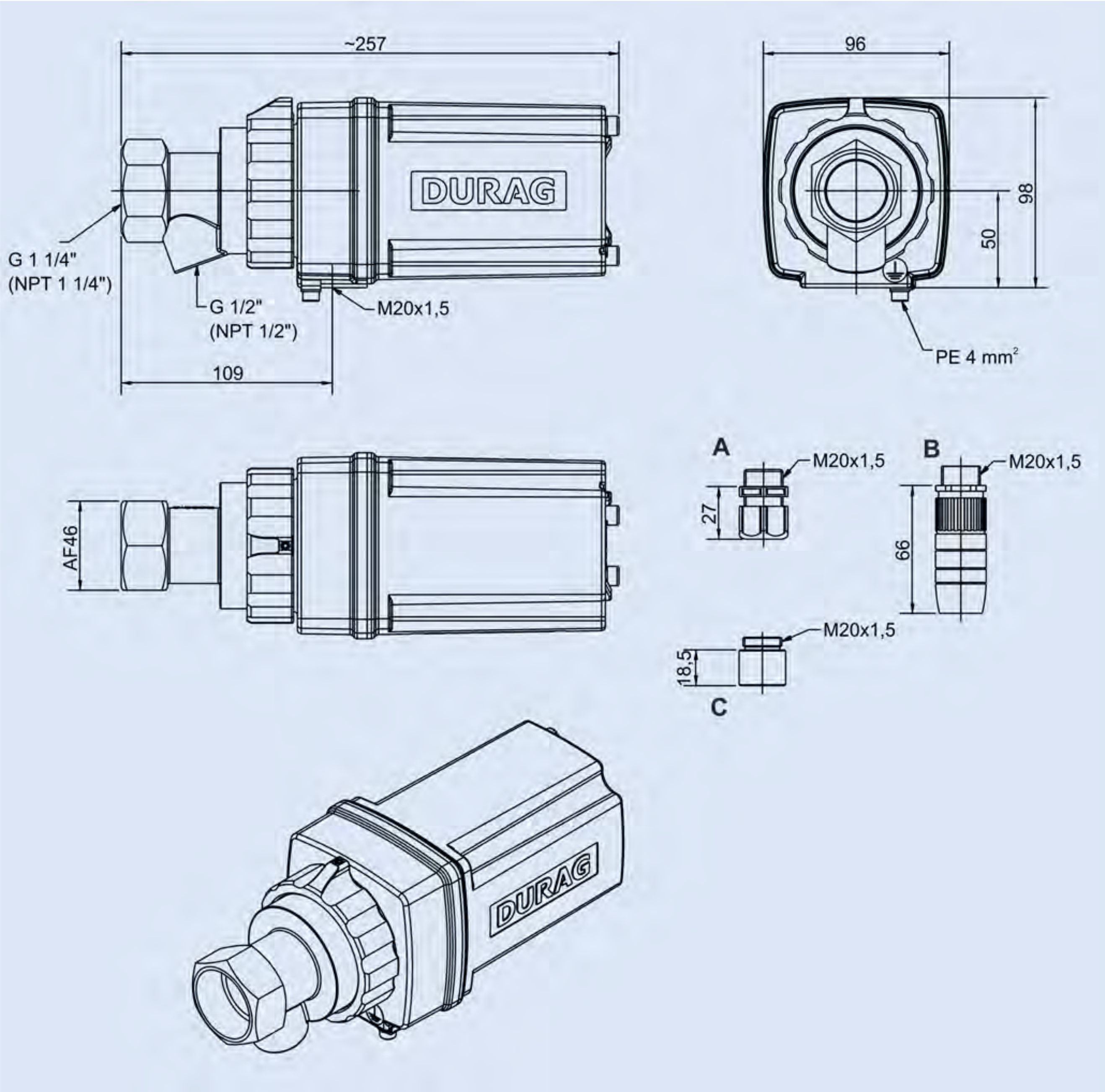


Fig. 19.2: Connection diagram for D-LX 110/710 with panel plug

20 Dimensional diagrams

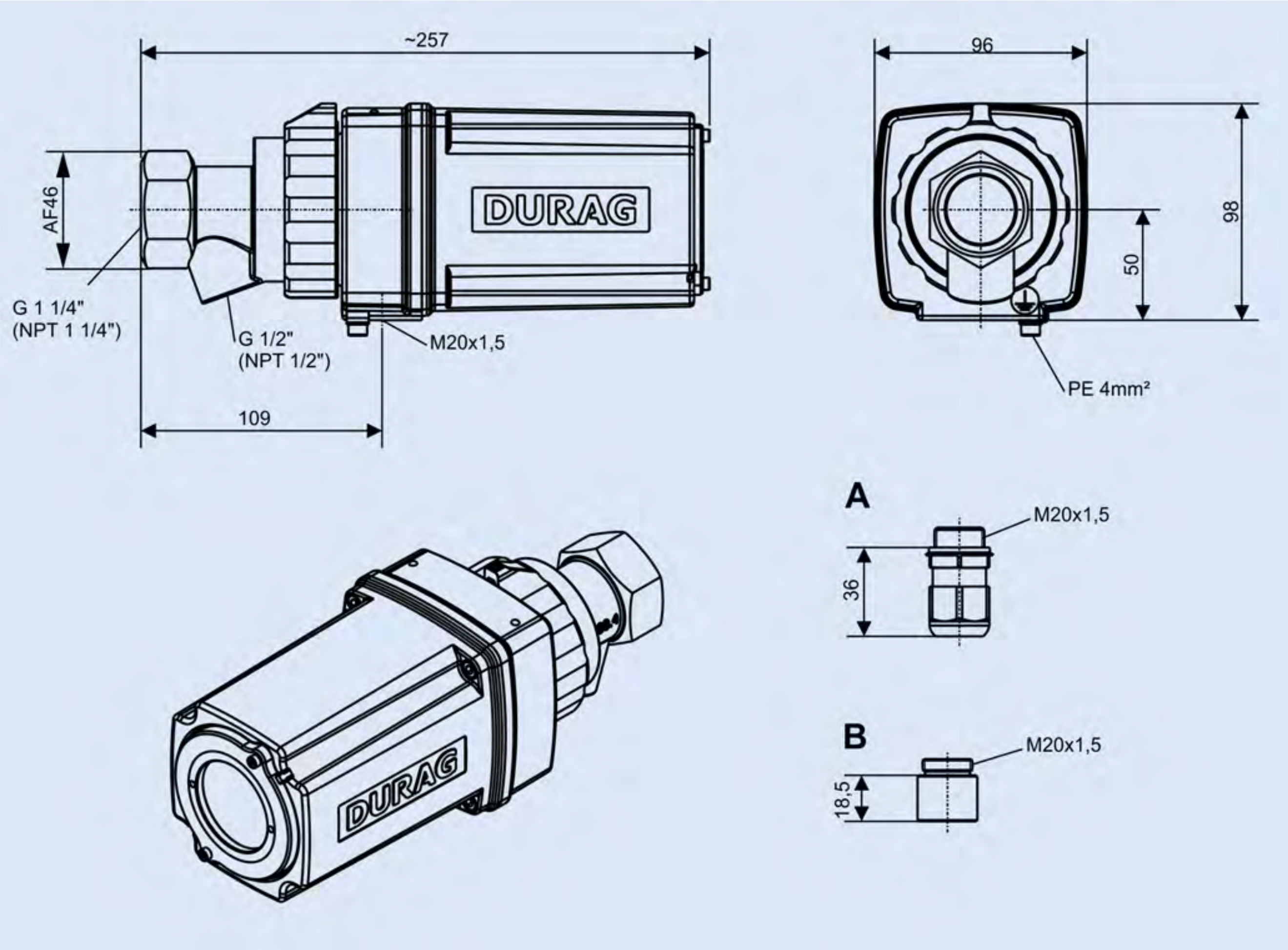
20.1 D-LX 110*M5*/0000* (no explosion protection)



A	Cable gland (plastic)	C	Adapter NPT ½"
B	Panel plug		

Fig. 20.1: The flame monitor can be fitted with either A, B or C for the electrical connection.

20.2 D-LX 110*M5*/86Ex* and D-LX 110*M5*/87Ex

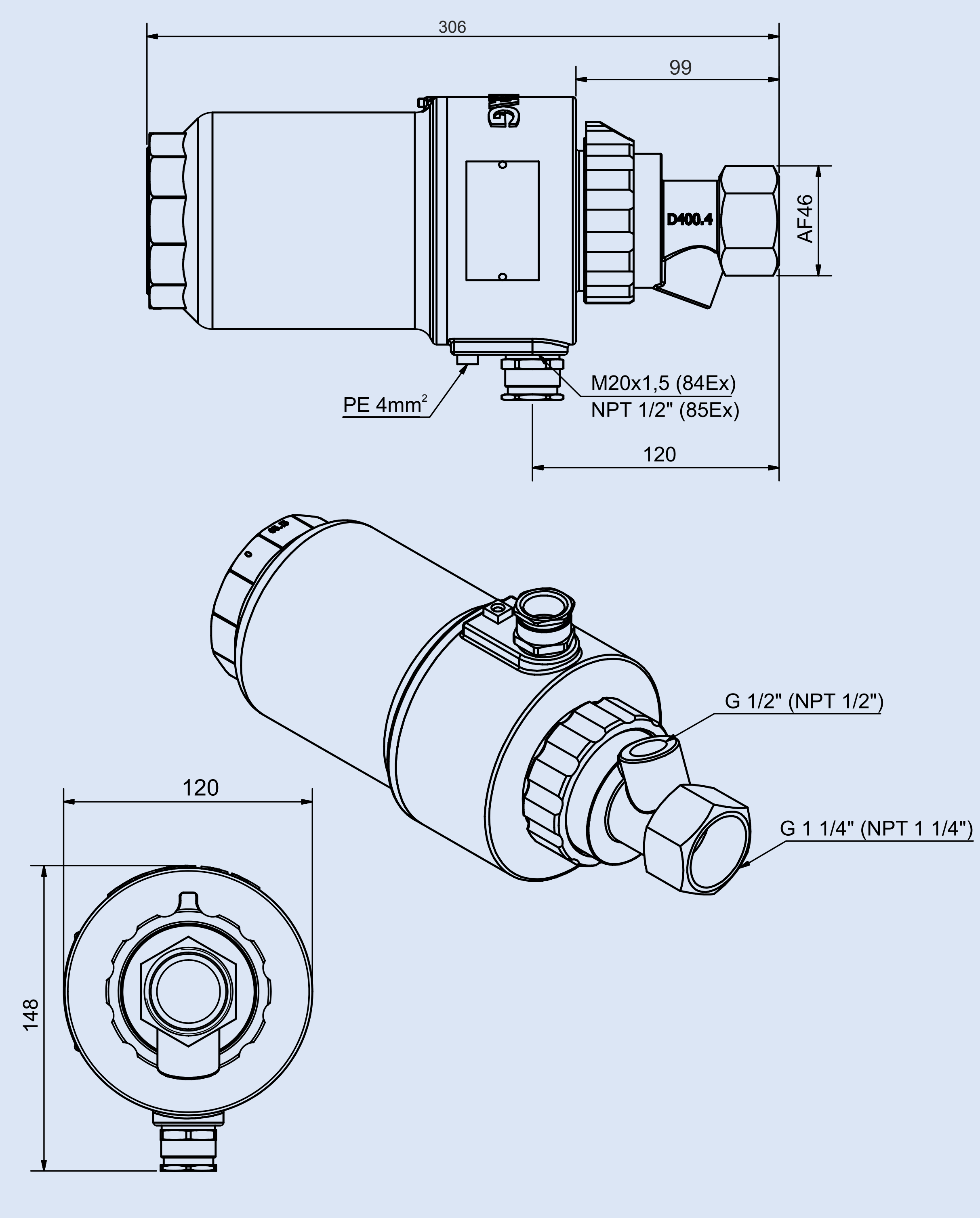


A	Cable gland
B	Adapter NPT ½"

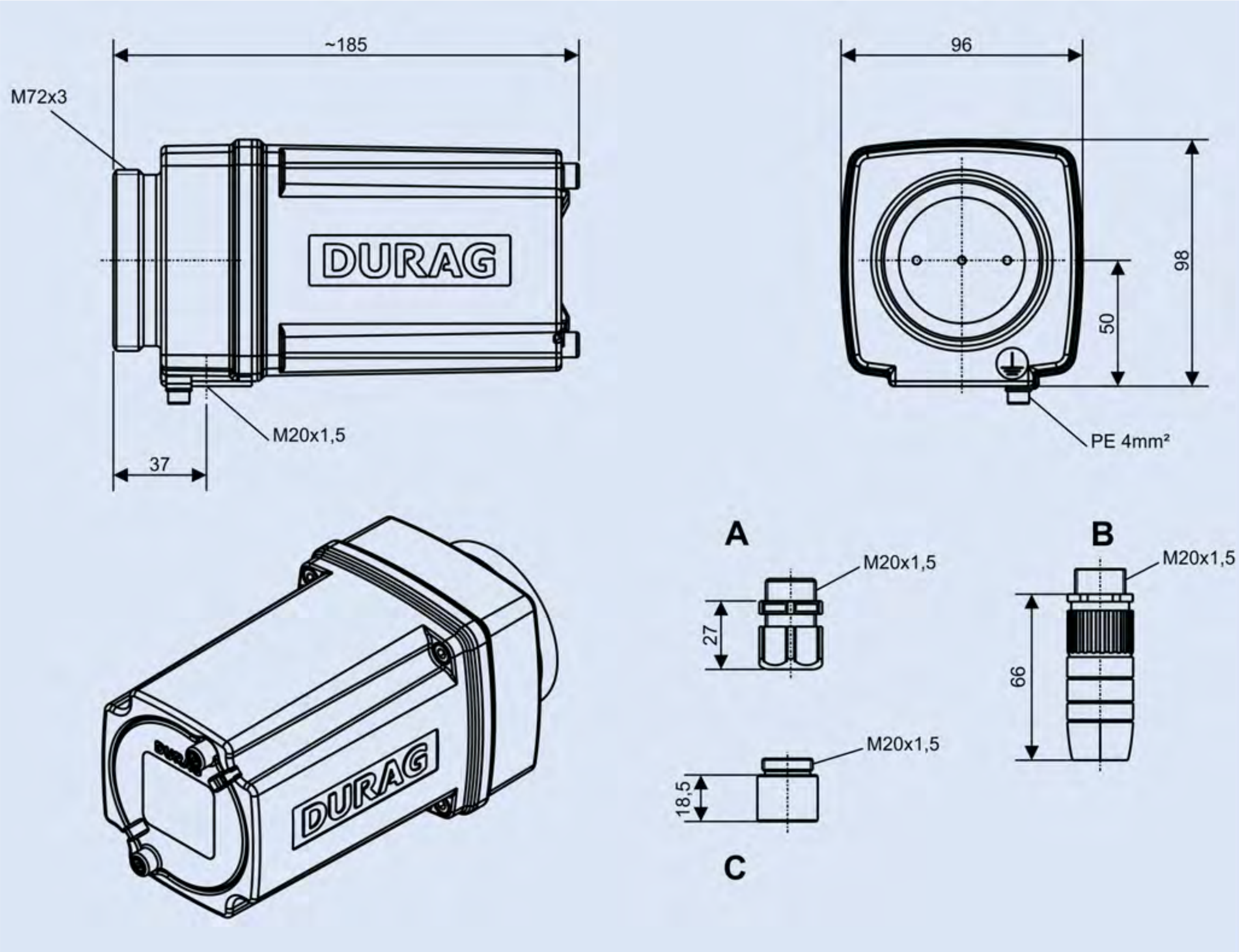
Fig. 20.2: The flame monitor can be fitted with either A or B for the electrical connection.

20.3 D-LX 110*M4*/84Ex* and D-LX 110*M4*/85Ex*

D-LX 110 M4



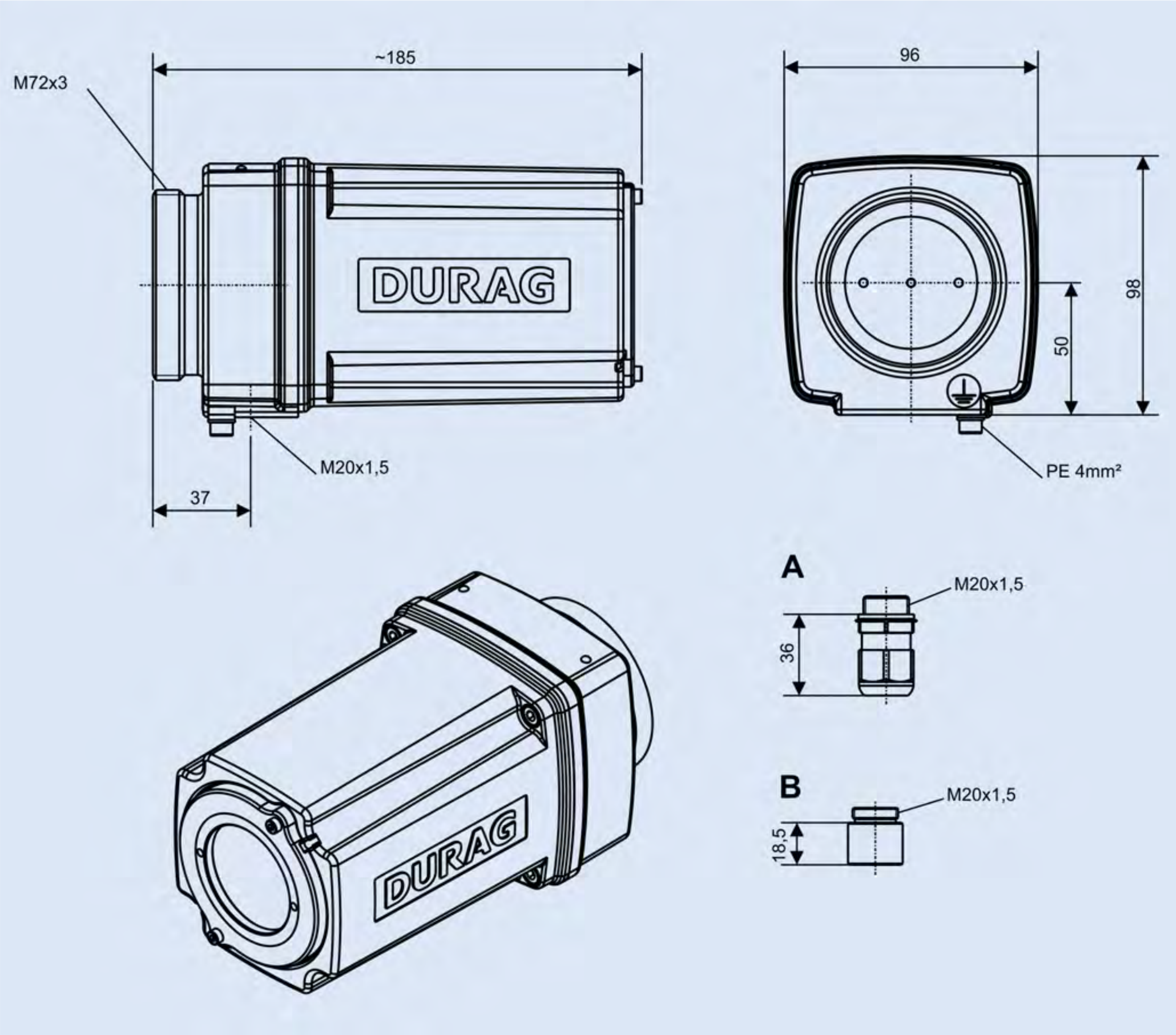
20.4 D-LX 710*M5*/0000* (no explosion protection)



A	Cable gland (plastic)	C	Adapter NPT ½"
B	Panel plug		

Fig. 20.3: The flame monitor can be fitted with either A, B or C for the electrical connection.

20.5 D-LX 710*M5*/86Ex* and D-LX 710*M5*/87Ex*

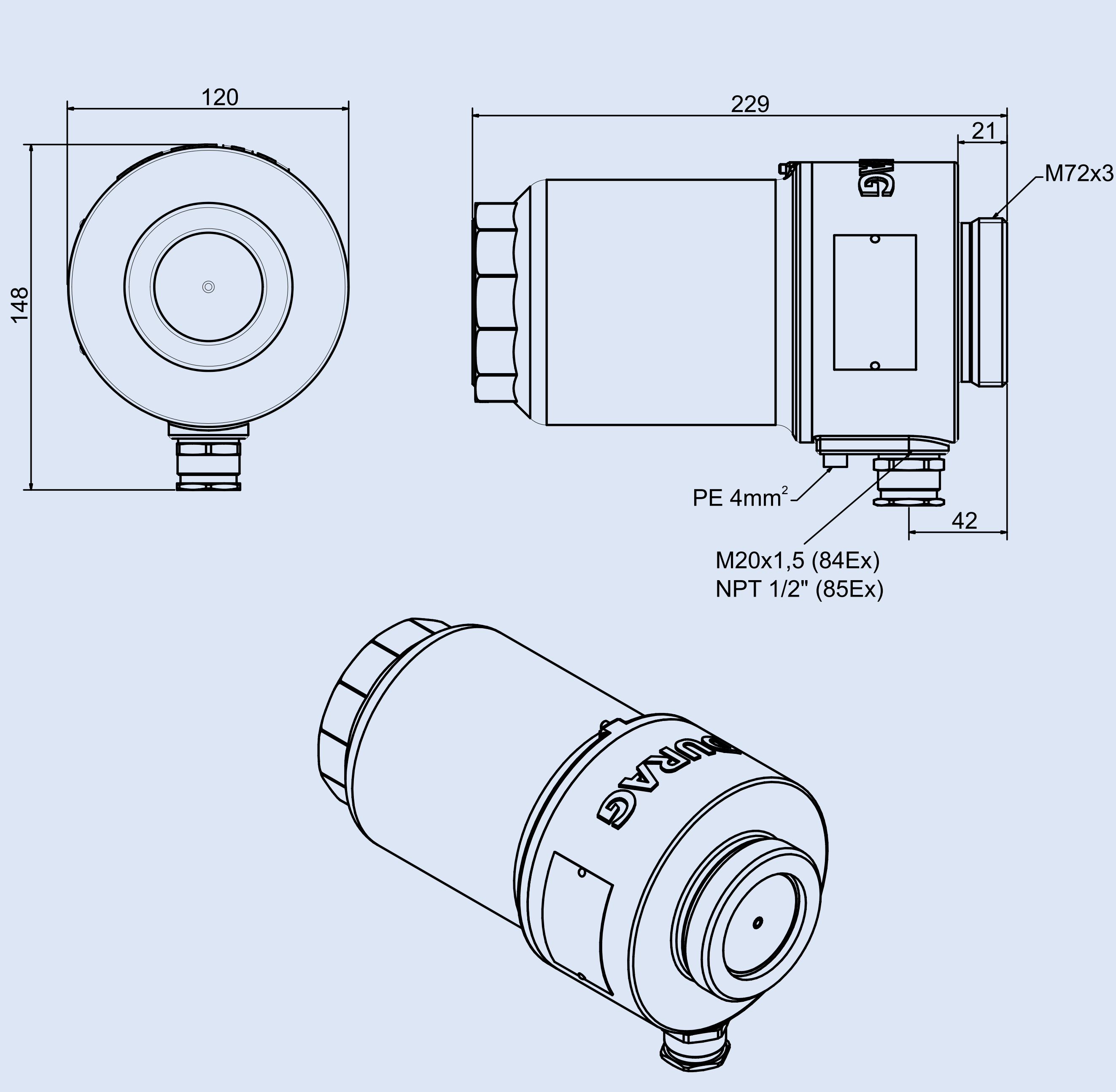


A	Cable gland
B	Adapter NPT ½"

Fig. 20.4: The flame monitor can be fitted with either A or B for the electrical connection.

20.6 D-LX 710*M4*/84Ex* and D-LX 710*M4*/85Ex*

D-LX 710 M4



21 Certificates

For many countries, national approvals are available for the flame monitor. However, not all types are available with all approvals. Detailed information on the approvals for your country can be obtained from your sales representative. A few certificates are listed below by means of example:

- DVGW certificate
- **IECEx** zone 1/21
- **IECEx** zone 2/22
- SIL3
- **FM** approval
- **FM** Class **I**, **II**, **III** Div. 1
- **FM** Class **I**, **II**, **III** Div. 2

21.1 According to gas appliance regulation (EU/2016/426)

CE 0085

DVGW
CERT

EC type examination certificate

EU-Baumusterprüfbescheinigung

CE-0085CS0429
Product Identification No.
Produkt-Identnummer

Field of Application ***Anwendungsbereich***	EC Gas Appliances Regulation (EU/2016/426) *EU-Gasgeräteverordnung (EU/2016/426)*
Owner of Certificate ***Zertifikatinhaber***	DURAG GmbH Kollaustraße 105, D-22453 Hamburg
Distributor ***Vertreiber***	DURAG GmbH Kollaustraße 105, D-22453 Hamburg
Product Category ***Produktart***	accessories for gas appliances/pressure equipment: Flame supervisor device (4131)
Product Description ***Produktbezeichnung***	flame detector device for permanent operation
Model ***Modell***	D-LX 110/710
Countries of Destination ***Bestimmungsländer***	European Union, CH, IS, NO
Test Reports ***Prüfberichte***	type testing: C-F 1529-01/17 from 27.10.2017 (TSG)
Test Basis ***Prüfgrundlagen***	EU/2016/426 A III B (09.03.2016) DIN EN 298 (01.11.2012)
Validity / File no. ***Gültigkeit / AZ***	21.04.2018 until 18.12.2027 / 17-0291-GEE

18.12.2017 Rie A-1/2
Date, Issued by, Sheet, Head of Certification Body
Datum, Bearbeiter, Blatt, Leiter der Zertifizierungsstelle

DVGW CERT GmbH is an accredited body by DAkkS according to DIN EN ISO/IEC 17065:2013 and notified by the government of the Federal Republic of Germany for certification of gas appliances under EC Regulation

DVGW CERT GmbH ist von der DAkkS nach DIN EN ISO/IEC 17065:2013 akkreditierte und von der Deutschen Bundesregierung benannte Stelle für die Zertifizierung von Gasgeräten gemäß EU-Verordnung EU/2016/426.

DAkkS
Deutsche
Akkreditierungsstelle
D-ZE-16028-01-01

DVGW CERT GmbH
Zertifizierungsstelle
Josef-Wirmer-Str. 1-3
53123 Bonn
Tel. +49 228 91 88 - 888
Fax +49 228 91 88 - 993
www.dvgw-cert.com
info@dvgw-cert.com

A-2/2 **CE-0085CS0429**

Elektrical Data *Elektrische Daten*	24 V DC

Type *Typ*	**Technical Data** *Technische Daten*	**Remarks** *Bemerkungen*
D-LX 110/710...	safety time: 1 s ambient temperature range: -40...+75 or -40...+70 (D-LX 110 UL...) °C	

Type Variation *Ausführungsvariante*	**Explanations** *Erläuterungen*
D-LX 110 UL...	with UV-sensor; spectral range: 185...260 nm
D-LX 110/710 UA...	with GaP-photodiode; spectral range: 190...570 nm
D-LX 110/710 UAF...	with GaP-photodiode with UV filter (UG 1); spectral range: 245...400 nm
D-LX 110/710 IG...	with GaAs-photodiode; spectral range: 800...1700 nm
D-LX 110/710 IS...	mit Si-photodiode; spectral range: 300...1100 nm

21.2 According to ATEX directive (2014/34/EU)

IBExU Institut für Sicherheitstechnik GmbH
An-Institut der TU Bergakademie Freiberg

[1] **EU-TYPE EXAMINATION CERTIFICATE - TRANSLATION**

[2] Equipment and protective systems intended for use in potentially explosive atmospheres, directive 2014/34/EU

[3] EU-Type Examination Certificate Number **IBExU11ATEX1050 X** | Issue 2

[4] Equipment: **Compact Flame Monitor**
Type: D-LX 200 *-*/*/84 Ex/* and D-LX 720 *-*/*/84 Ex/*
D-LX 201 *-*/*/84 Ex/* and D-LX 721 *-*/*/84 Ex/*
D-LX 110 *-*/*/84 Ex/* and D-LX 710 *-*/*/84 Ex/*

[5] Manufacturer: DURAG GmbH

[6] Address: Kollaustraße 105
22453 Hamburg
GERMANY

[7] This product and any acceptable variation thereto are specified in the schedule to this certificate and the documents therein referred to.

[8] IBExU Institut für Sicherheitstechnik GmbH, Notified Body number 0637 in accordance with Article 17 of Directive 2014/34/EU of the European Parliament and of the Council, dated 26 February 2014, certifies that this product has been found to comply with the essential health and safety requirements relating to the design and construction of products intended for use in potentially explosive atmospheres given in Annex II to the Directive.

The examination and test results are recorded in the confidential test report IB-18-3-0005.

[9] Compliance with the essential health and safety requirements has been assured by compliance with:
EN 60079-0:2012+A11:2013 **EN 60079-1:2014** **EN 60079-31:2014**
Except in respect of those requirements listed at item [18] of the annex.

[10] If the sign "X" is placed after the certificate number, it indicates that the product is subject to the specific conditions of use specified in the schedule to this certificate.

[11] This EU-type examination certificate relates only to the design and construction of the specified product. Further requirements of the Directive apply to the manufacturing process and supply of this product. These are not covered by this certificate.

[12] The marking of the product shall include the following:

Ex II 2G Ex db IIC T6 or T5 Gb
Ex II 2D Ex tb IIIC T85 °C or T100 °C Db
$-40\ °C \le T_a \le 70\ °C \ldots 85\ °C$

IBExU Institut für Sicherheitstechnik GmbH
Fuchsmühlenweg 7
09599 Freiberg, GERMANY

Phone: +49 (0)3731 3805-0
Fax: +49 (0)3731 3805-10

Certificates without seal and signature are not valid. Certificates may only be duplicated completely and unchanged. In case of dispute, the German text shall prevail.

By order

A. Henker
Dipl.-Ing. (FH) Henker

IBExU Institut für Sicherheitstechnik GmbH
- Seal -
(Notified Body number 0637)

Freiberg, 01 October 2018

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IBExU Institut für Sicherheitstechnik GmbH
An-Institut der TU Bergakademie Freiberg

[13] **Schedule**

[14] **Certificate Number IBExU11ATEX1050 X** | Issue 2

[15] **Description of product**

The Compact Flame Monitors type D-LX 200 *-*/*/84 Ex/* and D-LX 720 *-*/*/84 Ex/*, D-LX 201 *-*/*/84 Ex/* and D-LX 721 *-*/*/84 Ex/* as well as D-LX 110 *-*/*/84 Ex/* and D-LX 710 *-*/*/84 Ex/* are used in combustion systems. The equipment serves as switching and measuring devices.
The flameproof enclosure from aluminium casting or stainless steel (only type D-LX 200/201/110) consists of the cover with sight glass and the middle part with glass lens (type D-LX 200/201/110) or optical fibre (type D-LX 720/721/710). The electric connection is carried out with direct cable entry.

Technical data

- Operating voltage:	24 V DC ± 20 %
- Dissipation power:	max. 3.7 W (Type D-LX 20*/72*) max. 4.0 W (Type D-LX 110/710)
- Relay output:	24 V DC, max. 2 x 0.5 A or 2 A
- Ambient temperature range:	-40 °C up to +70 °C (T6 or T85 °C) -40 °C up to +75 °C (T5 or T100 °C, type D-LX 110/710) -40 °C up to +85 °C (T5 or T100 °C, type D-LX 20*/72*)
- Degree of protection:	IP66 (according to EN 60529)

[16] **Test report**

The test results are recorded in the confidential test report IB-18-3-0005 of 01 October 2018. The test documents are part of the test report and they are listed there.

Summary of the test results

The Compact Flame Monitors type D-LX 200 *-*/*/84 Ex/* and D-LX 720 *-*/*/84 Ex/*, D-LX 201 *-*/*/84 Ex/* and D-LX 721 *-*/*/84 Ex/* as well as D-LX 110 *-*/*/84 Ex/* and D-LX 710 *-*/*/84 Ex/* fulfil the requirements of explosion protection for equipment of Group II, Category 2G, type of protection flameproof enclosure „db" and Category 2D, type of protection dust ignition protection by enclosure „tb".

Variations compared to issue 1 of this certificate:

- Addition of the types D-LX 110 *-*/*/84 Ex/* and D-LX 710 *-*/*/84 Ex/*

[17] **Special conditions for use**

- Repairs of the flameproof joints must be made in compliance with the constructive specifications provided by the manufacturer. Repairs must not be made on the basis of values specified in tables 2 and 3 of EN 60079-1.
- The Compact Flame Monitor type D-LX *** *-*/*/84 Ex/MCG is optionally equipped by the manufacturer with cable gland and connecting cable. This cable gland may only be used for fixed installation. The free cable end must be connected either outside the potentially explosive atmosphere or in an equipment certified for the respective category. The length of the cable must be at minimum 0.5 m.
- At own selection of the cable glands and the connection cable the corresponding requirements in EN 60079-14, Paragraph 10.6 have to be noticed. The cable glands have to be suitable and certified accordingly. Unused openings for cable entries have to be closed durably with suitable screw plugs, which are confirmed for explosion protection according to EN 60079-1, 11.9.
- At maximum ambient temperatures >70 °C, only cable glands and connecting cables suitable for the temperatures specified in the operating instructions should be used.

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IBExU Institut für Sicherheitstechnik GmbH

An-Institut der TU Bergakademie Freiberg

[18] **Essential health and safety requirements**

In addition to the essential health and safety requirements (EHSRs) covered by the standards listed at item [9], the following are considered relevant to this product, and conformity is demonstrated in the test report:

- not applicable -

[19] **Drawings and documents**

The documents are listed in the test report.

By order

Freiberg, 01 October 2018

A. Henker

Dipl.-Ing. (FH) Henker

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IBExU Institut für Sicherheitstechnik GmbH
An-Institut der TU Bergakademie Freiberg

[1] **TYPE EXAMINATION CERTIFICATE** - Translation

[2] for non-electrical products of equipment-groups I and II,
equipment-categories M2 and 2 plus products of equipment-category 3

[3] Type examination certificate number **IBExU18ATEXB016** | Issue 0

[4] Product: **Compact Flame Monitor**
Type: D-LX 110*/87 Ex and D-LX 710*/87 Ex

[5] Manufacturer: DURAG GmbH

[6] Address: Kollaustraße 105
22453 Hamburg
GERMANY

[7] This product and any acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

[8] IBExU Institut für Sicherheitstechnik GmbH certifies that this product has been found to comply with the essential health and safety requirements relating to the design and construction of products intended for use in potentially explosive atmospheres given in Annex II to Directive 2014/34/EU of the European Parliament and of the Council, dated 26 February 2014.

The examination and test results are recorded in the confidential test report IB-18-3-0013.

[9] Compliance with the essential health and safety requirements has been assured by compliance with:
EN 60079-0:2018, EN 60079-7:2015, EN 60079-15:2018 and EN 60079-31:2014
except in respect of those requirements listed at item [18] of the schedule.

[10] If the sign "X" or "U" is placed after the certificate number, it indicates that the product is subject to the specific conditions of use specified in the schedule to this certificate.

[11] This type examination certificate relates only to the design of the specified equipment and not to specific items of equipment subsequently manufactured or supplied.

[12] The marking of the product shall include the following:

Ex II 3G Ex ec nC IIC T6…T4 Gc
Ex II 3D Ex tc IIIC T80 °C…T130 °C Dc
-40 °C ≤ T_{amb} ≤ +40 °C or +70 °C

IBExU Institut für Sicherheitstechnik GmbH
Fuchsmühlenweg 7
09599 Freiberg, GERMANY

Tel: + 49 (0) 37 31 / 38 05 0
Fax: + 49 (0) 37 31 / 38 05 10

By order

IBExU
Institut für Sicherheitstechnik GmbH
Fuchsmühlenweg 7
09599 Freiberg/Sachsen
Telefon (03731) 3805-0
Telefax (03731) 38 05 10

Dipl.-Ing. (FH) Henker - Stamp - Freiberg, 2018-11-12

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IBExU Institut für Sicherheitstechnik GmbH
An-Institut der TU Bergakademie Freiberg

[13] **Schedule**

[14] **Certificate number IBExU18ATEXB016** | Issue 0

[15] **Description of product**
Flame monitors are safety devices and are designed to ensure the safe operation of a furnace plant. The D-LX * Flame monitor consists of a control unit and an optical flame scanner. The flame monitor is suitable for monitoring flames from a variety of fuels and combustion techniques, particularly in single burner applications. A photo element, evaluation electronics and a switchgear are housed together in an aluminium enclosure.

Technical data

Ambient temperature range:	-40 °C up to +40 °C (T6 or T80 °C)
Ambient temperature range:	-40 °C up to +70 °C (T4 or T130 °C)
Degree of protection:	IP66 according to EN 60529

<u>Electrical data</u>

Operating voltage:	24 ± 20 % V DC
Nominal current:	150 mA (at 24 V)
Relay output:	max. 250 V AC or 24 V DC max. 2 A

[16] **Test report**
The test results are recorded in the confidential test report IB-18-3-0013 of 2018-11-12.

The test documents are part of the test report and they are listed there.

Summary of the test results
The Compact Flame Monitors type D-LX 110*/87 Ex und D-LX 710*/87 Ex fulfil the requirements of explosion protection for apparatus of the Equipment Group II and Category 3 G in type of protection increased safety "ec" in combination with sealed devices "nC" as well as the requirements of the category 3D in type of protection Protection by enclosure "tc".

[17] **Specific conditions of use**
None

[18] **Essential health and safety requirements**
In addition to the essential health and safety requirements (EHSRs) covered by the standards listed at item [9], the following are considered relevant to this product, and conformity is demonstrated in the test report:
None

[19] **Drawings and Documents**
The documents are listed in the test report.

IBExU Institut für Sicherheitstechnik GmbH
Fuchsmühlenweg 7
09599 Freiberg, GERMANY

By order

A. Henker

Dipl.-Ing. [FH] Henker

Freiberg, 2018-11-12

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21.3 IECEx-Certificates

IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEC Certification Scheme for Explosive Atmospheres
for rules and details of the IECEx Scheme visit www.iecex.com

Certificate No.:	IECEx IBE 11.0012X	Issue No: 3	Certificate history:
Status:	Current		Issue No. 3 (2018-10-19)
			Issue No. 2 (2017-12-07)
		Page 1 of 4	Issue No. 1 (2013-10-02)
Date of Issue:	2018-10-19		Issue No. 0 (2011-11-02)

Applicant: DURAG GmbH
Kollaustraße 105
22453 Hamburg
Germany

Equipment: Compact Flame Monitor
Optional accessory: *types D-LX 200 *-*/*/84 Ex/* and D-LX 720 *-*/*/84 Ex/*, D-LX 201 *-*/*/84 Ex/* and D-LX 721 *-*/*/84 Ex/*, D-LX 110 *-*/*/84 Ex/* and D-LX 710 *-*/*/84 Ex/**

Type of Protection: Flameproof enclosure "db"; Protection by enclosure "tb"

Marking:
Ex db IIC T6 ... T5 Gb
Ex tb IIIC T85 °C ...100 °C Db
-40°C ≤ Ta ≤ 70 °C ... 85 °C

Approved for issue on behalf of the IECEx Certification Body: Dipl.-Ing. Alexander Henker

Position: Deputy Head of Certification Body

Signature: (for printed version) A. Henker

Date: 2018-10-19

1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
3. The Status and authenticity of this certificate may be verified by visiting the Official IECEx Website.

Certificate issued by:

IBExU Institut für Sicherheitstechnik GmbH
Certification Body
Fuchsmühlenweg 7
09599 Freiberg
Germany

IECEx Certificate of Conformity

Certificate No:	IECEx IBE 11.0012X	Issue No: 3
Date of Issue:	**2018-10-19**	Page 2 of 4
Manufacturer:	**DURAG GmbH** Kollaustraße 105 22453 Hamburg **Germany**	

Additional Manufacturing location(s):

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended.

STANDARDS:

The apparatus and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards:

IEC 60079-0 : 2011 Edition:6.0	Explosive atmospheres - Part 0: General requirements
IEC 60079-1 : 2014-06 Edition:7.0	Explosive atmospheres - Part 1: Equipment protection by flameproof enclosures "d"
IEC 60079-31 : 2013 Edition:2	Explosive atmospheres - Part 31: Equipment dust ignition protection by enclosure "t"

*This Certificate **does not** indicate compliance with electrical safety and performance requirements other than those expressly included in the Standards listed above.*

TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in

Test Report:

DE/IBE/ExTR11.0012/00 DE/IBE/ExTR11.0012/01 DE/IBE/ExTR11.0012/02
DE/IBE/ExTR11.0012/03

Quality Assessment Report:

DE/TUN/QAR11.0010/08

IECEx Certificate of Conformity

Certificate No:	IECEx IBE 11.0012X	Issue No: 3
Date of Issue:	2018-10-19	Page 3 of 4

Schedule

EQUIPMENT:

Equipment and systems covered by this certificate are as follows:

The Compact Flame Monitors type D-LX 200 *-*/ */84 Ex/ * and D-LX 720 *-*/ */84 Ex/ *, D-LX 201 *-*/ */84 Ex/ * and D-LX 721 *-*/ */84 Ex/ * as well as D-LX 110 *-*/ */84 Ex/ * and D-LX 710 *-*/ */84 Ex/ * are used in combustion systems. The equipment serves as switching and measuring devices.

The flameproof enclosure from aluminium casting or stainless steel (only type D-LX 200/201/110) consists of the cover with sight glass and the middle part with glass lens (type D-LX 200/201/110) or optical fibre (type D-LX 720/721/710). The electric connection is carried out with direct cable entry.

Technical data

Operating voltage:	24 V DC ± 20 %
Dissipation power:	max. 3.7 W (type D-LX 20*/72*)
	max. 4.0 W (type D-LX 110/710)
Relay output:	24 V DC, max. 2 x 0.5 A or 2 A
Ambient temperature range:	-40 °C up to +70 °C (T6 or T85 °C)
	-40 °C up to +75 °C (T5 or T100 °C, type D-LX 110/710)
	-40 °C up to +85 °C (T5 or T100 °C, type D-LX 20*/72*)
Degree of protection:	IP66 (according to IEC 60529)

SPECIFIC CONDITIONS OF USE: YES as shown below:

- Repairs of the flameproof joints must be made in compliance with the constructive specifications provided by the manufacturer. Repairs must not be made on the basis of values specified in tables 2 and 3 of IEC 60079-1.
- The Compact Flame Monitor may be equipped by the manufacturer with cable gland and connecting cable. This cable gland has to be used only for fixed installation. The connection of the free cable end must be carried out either outside the potentially explosive atmosphere or in an equipment certified for the appropriate category. The length of the cable must be minimum 0.5 m.
- At own selection of the cable glands and the connection cable the corresponding requirements in IEC 60079-14, Paragraph 10.6.2 have to be noticed. The cable glands have to be suitable and certified accordingly. Unused openings for cable entries have to be closed durably with suitable screw plugs, which are confirmed for explosion protection according to IEC 60079-1 and IEC 60079-31.
- At maximum ambient temperatures >60 °C, only cable glands and connecting cables suitable for the temperatures specified in the operating instructions should be used.

IECEx Certificate of Conformity

Certificate No:	IECEx IBE 11.0012X	Issue No: 3
Date of Issue:	**2018-10-19**	Page 4 of 4

DETAILS OF CERTIFICATE CHANGES (for issues 1 and above):

Addition of the types D-LX 110 *-*/ */84 Ex/ * and D-LX 710 *-*/ */84 Ex/ *

IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEC Certification Scheme for Explosive Atmospheres

for rules and details of the IECEx Scheme visit www.iecex.com

Certificate No.: IECEx IBE 18.0024

Issue No: 0

Certificate history:
Issue No. 0 (2018-11-12)

Status: **Current**

Page 1 of 3

Date of Issue: **2018-11-12**

Applicant: **DURAG GmbH**
Kollaustraße 105
22453 Hamburg
Germany

Equipment: **Compact Flame Monitor type D-LX 110*/87 Ex and D-LX 710*/87 Ex**

Optional accessory:

Type of Protection: **Increased safety "e" in combination with type of protection "n" and protection by enclosure "t"**

Marking:

Ex ec nC IIC T6...T4 Gc

Ex tc IIIC T80 °C...T130 °C Dc

-40 °C ≤ T_{amb} ≤ +40 °C or +70 °C

Approved for issue on behalf of the IECEx Certification Body: Dipl.-Ing. Alexander Henker

Position: Deputy Head of Certification Body

Signature:
(for printed version) A. Henker

Date: 2018-11-12

1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
3. The Status and authenticity of this certificate may be verified by visiting the Official IECEx Website.

Certificate issued by:

IBExU Institut für Sicherheitstechnik GmbH
Certification Body
Fuchsmühlenweg 7
09599 Freiberg
Germany

IBExU

IECEx Certificate of Conformity

Certificate No:	IECEx IBE 18.0024	Issue No: 0
Date of Issue:	**2018-11-12**	Page 2 of 3
Manufacturer:	**DURAG GmbH** Kollaustraße 105 22453 Hamburg **Germany**	

Additional Manufacturing location(s):

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended.

STANDARDS:

The apparatus and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards:

IEC 60079-0 : 2017 Edition:7.0	Explosive atmospheres - Part 0: Equipment - General requirements
IEC 60079-15 : 2017 Edition:5.0	Explosive atmospheres - Part 15: Equipment protection by type of protection "n"
IEC 60079-31 : 2013 Edition:2	Explosive atmospheres - Part 31: Equipment dust ignition protection by enclosure "t"
IEC 60079-7 : 2015 Edition:5.0	Explosive atmospheres – Part 7: Equipment protection by increased safety "e"

This Certificate ***does not*** *indicate compliance with electrical safety and performance requirements other than those expressly included in the Standards listed above.*

TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in

Test Report:

DE/IBE/ExTR18.0013/00

Quality Assessment Report:

DE/TUN/QAR11.0010/08

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Schedule

EQUIPMENT:

Equipment and systems covered by this certificate are as follows:

Flame monitors are safety devices and are designed to ensure the safe operation of a furnace plant. The D-LX * All-In-One Flame monitor consists of a control unit and an optical flame scanner. The flame monitor is suitable for monitoring flames from a variety of fuels and combustion techniques, particularly in single burner applications.

Technical data:

operating voltage:	24 V DC ± 20 %
nominal current:	150 mA (at 24 V)
relay output:	maximum 250 V, max. 2 A
degree of protection according to IEC 60529:	IP66
ambient temperature range:	-40 °C up to +40 °C (T6 or T80 °C) -40 °C up to +70 °C (T4 or T130 °C)

SPECIFIC CONDITIONS OF USE: NO

22 Glossary

Background radiation

The radiation of static emitters such as glowing boiler components can be modulated by combustion air currents or flue gas vapours. If the measured radiation is in the range of the flame spectrum and the usual flicker frequency range (approx. 10 to 200 Hz) for a flame, then the flame monitor may incorrectly detect a flame signal. Radiation from the flames in other burners can also be incorrectly detected as a flame signal.

FFDT

abbreviation for: flame failure detection time, see flame failure detection time

Flame failure detection time

The flame failure detection time (FFDT) is the flame monitor's response time to a failure of the flame signal and the resulting deactivation of the relay contact for the Flame ON signal. The FFDT was formerly known as the safety time and is referred to as the "Detektionszeit bei Flammenausfall" in German.

Flame radiation

is the visible and invisible (IR/UV) light emitted by the flame.

Flame signal

In the photo diode, the flame generates an (analogue) current, which is then processed in the analogue amplifier and the analogue filters. This flame signal (AC/DC) is transferred to the microcontroller inputs for further digital processing.

Safety time

See flame failure detection time

Switching threshold

The switching threshold determines whether the magnitude of the existing flame signal should result in a Flame ON or Flame OFF signal. Only flame signals with pulse rates higher than the switching threshold can be rated as Flame ON.

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DURAG